# Signals and Systems

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# Course Introduction

This course is a professional foundation course that examines the basic concepts and basic analytical methods of signals and systems. Preliminary understanding of how to establish a **mathematical model** of signals and systems, and after mathematical analysis and solution, the results are physically explained and given **physical meaning**.

## Textbook

Simon Haykin and Barry Van Veen, Signals and Systems (Second Edition), John Wiley & Sons, 2002, 9780471164746.

## Reference book

Alan V. Oppenheim, Signals and Systems (Second Edition), Prentice-Hall, 1996.

# Course Introduction

## Why Study Signals & Systems in the AI Era?

**DeepSeek：Signals & Systems (S&S) is the Foundational Language Connecting Physics to AI.**

### 1. AI's Data is Signals
•**Computer Vision:** Images = 2D Spatial Signals
•**Speech/Audio:** Sound = 1D Time-Based Signals
•**Sensor Data (IoT, AV):** LiDAR, Radar = Continuous Signal Streams
•**S&S Provides the Tools** to represent, analyze, and process this real-world data.

### 2. It's the Math Behind Core AI Architectures
•**CNNs (Convolutional Neural Networks):** Their core operation, **convolution**, is a fundamental S&S concept for filtering and feature extraction.
•**RNNs/LSTMs:** Process sequential data (time-series signals) using concepts of system memory and feedback.

### 3. The Critical Analog-to-Digital Bridge
•**Sampling Theorem (Nyquist):** Prevents **aliasing** (data distortion) when converting real-world analog data for AI models.
•**Digital Filter Design:** Essential for preprocessing data (noise removal, feature enhancement) before it reaches the model.

### 4. System-Level Thinking for Real-World AI
•**Beyond the Model:** AI is part of a larger **system** (e.g., a robot, a self-driving car).
•**Control Theory & Feedback:** S&S provides concepts for stability, response time, and latency crucial for real-time, reliable deployment.

**Without S&S:** You are an **API User**. You can apply tools but may struggle to debug, innovate, or handle complex, real-world data effectively.
**With S&S:** You become a **System Architect**. You understand *why* models work, can design better systems, and push the boundaries of what's possible.
**It's not outdated—it's essential for building robust, efficient, and innovative AI.**

# Assessment

| Assessment items | Percentage | Description |
|---|---|---|
| Final Exam | 80% | Scope of assessment: all contents. Closed-book exam. |
| Class Participation | 20% | The score is calculated according to the percentage of the number of answers to the number of questions initiated by the teacher to participate in the *Rain Class*, regardless of whether it is right or wrong. |

## Contents of this lecture



## Goals of this lecture

1. Understand how signals are classified
2. Understand the characteristics and physical significance of typical signals;
3. Proficient in signal calculation, be able to correctly draw the waveform after time shift, reverse fold and scale transformation step by step;
4. Familiarize yourself with the properties, effects, and interrelationships of exotic signals, especially step and impulse functions.

# 1.1 Introduction

Message: A symbol to be transmitted in a way agreed in advance by both the sender and receiver, such as language, text, images, data, etc.

Information: Unknown content obtained from the received message.

Signal: A signal is a **variation** of a measurable quantity that carries **information** about the behaviour of a system. Examples: human voice, room temperature controlled by a thermostat system, position, speed, and acceleration of an aircraft, force measured with force sensors in robotic systems, digital photographs, music recording etc.

Electrical signal: The current or voltage that changes corresponding to the message (language, text, image, data), or the charge on the capacitor, the magnetic flux in the inductor, etc.

# Signal Processing

- We interpret signals to make **decisions**.
- Signals allow us to see, hear, feel and act.
- Signal analysis allows us to **understand** or interpret an aspect of a signal which is *significant but not obvious on first inspection*.
- **Signal processing** is the science of analyzing, modifying, or synthesizing signals.

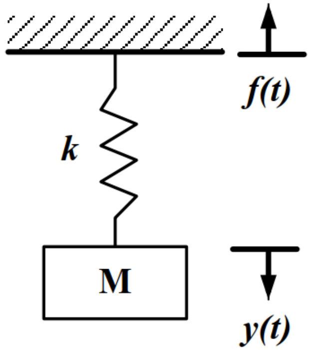
**Please name at least 2 applications of signal processing.**

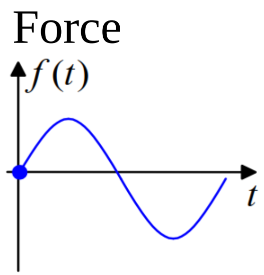
# 1.1 Introduction

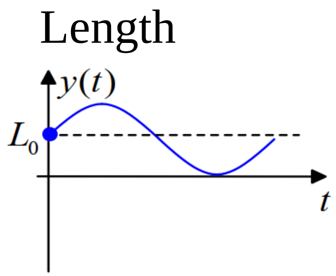
**System: A set of things that are connected to each other in a specific function.**


- Physical systems: Circuit systems, communication systems, automatic control systems, mechanical systems, optical systems, etc.;
- Non-physical systems: biological systems, political system systems, economic structure systems, transportation systems, meteorological systems, etc.

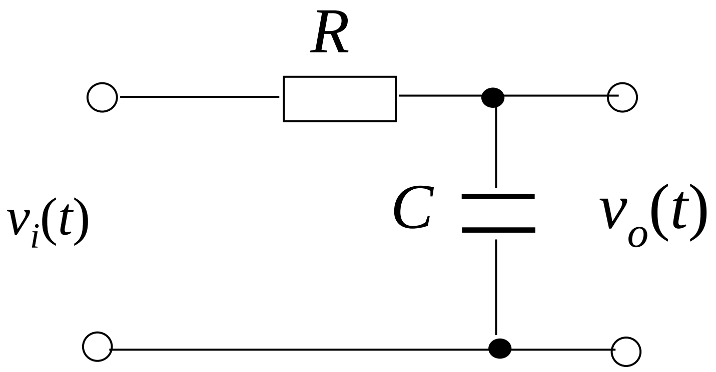
**Spring system**



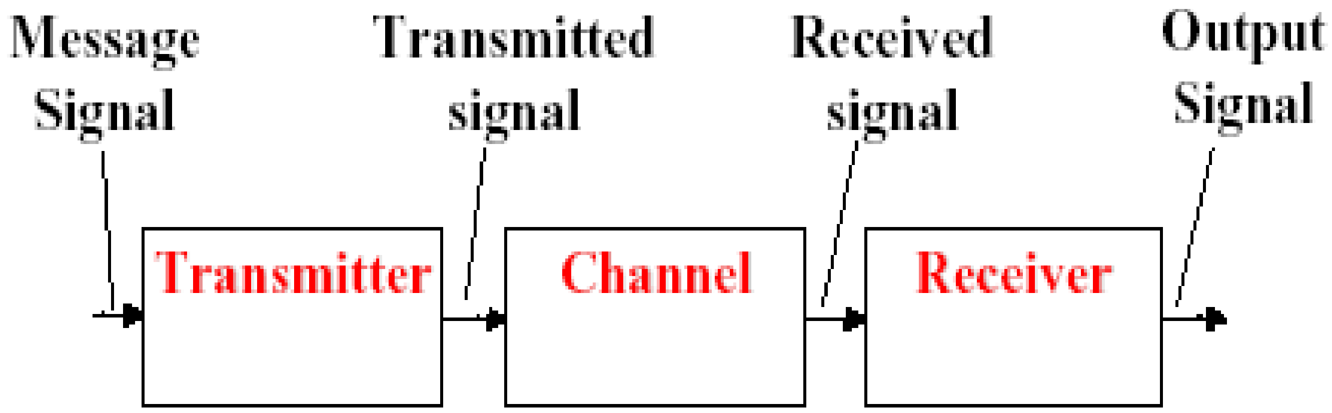



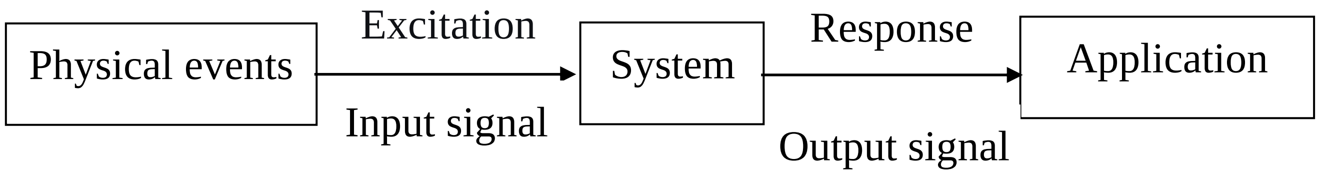
- **Electrical Circuit System**




- **Communication system**



# 1.1 Introduction




For a system, the input signal is often called **excitation**, and the output signal is often called the **response**.

Problems solved by this course in Signals and Systems:

1) What kind of response does the incentive system produce?

2) What kind of system is designed for a given excitation and required response?

3) What kind of incentives are needed to facilitate system analysis and the required response?

Excitation + system → response

Excitation → system ← response

Excitation ← System + Response

Note: Signals and systems are closely related and complement each other. Without the signal, the system loses its meaning.

## 1.2.1 Classification of Signals

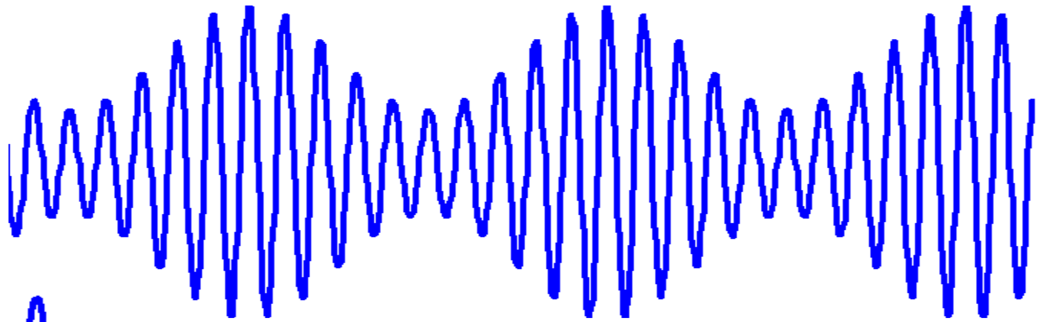
### 1. Deterministic Signals and Random Signals

Deterministic signal - for a certain moment, the signal has a definite value corresponding to it.

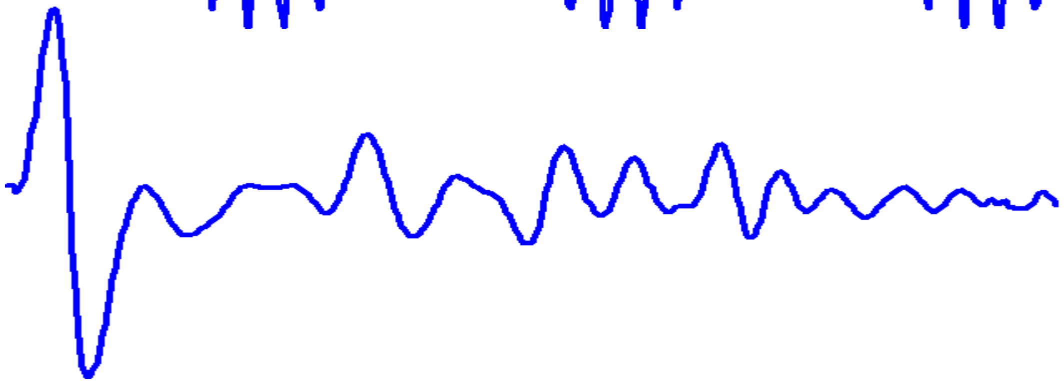
Random signals - unpredictable signals such as noise.

Deterministic signals

$S(t) = (2 + \cos\omega t)\cos(10\omega t)$



Random signal



## 2. Periodic Signals and Non-Periodic Signals

Periodic signal: It starts and starts at a certain time interval, and it is a signal without beginning or end. A signal with a basic period of $T$- $f(t)=f(t+T)$ for all $t$.

Non-periodic signal: A signal that does not meet the characteristics of the cycle in time.

The spreading code of direct-sequence spread spectrum is

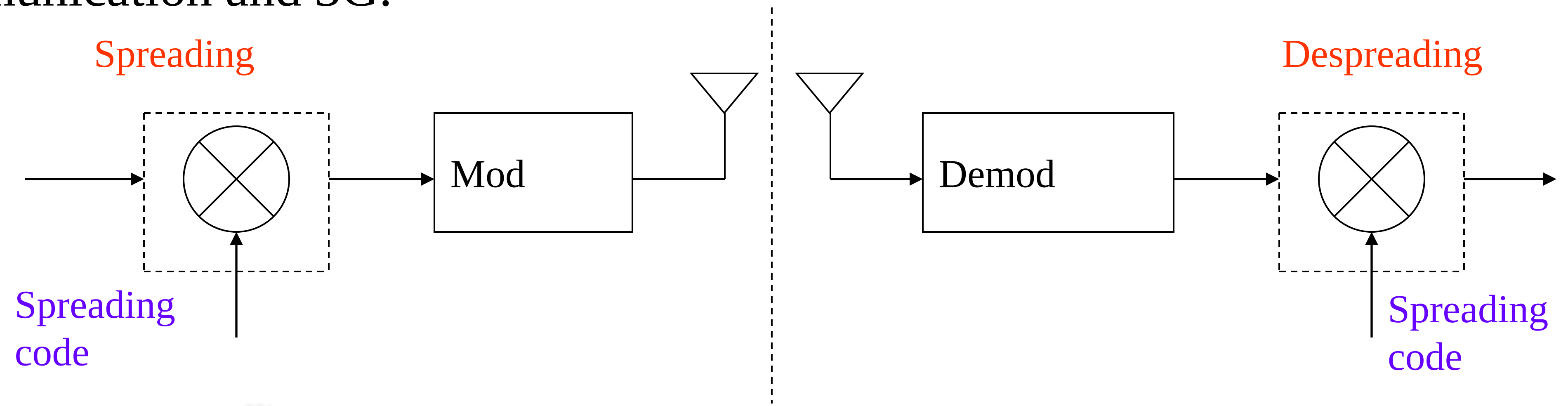
A A periodic signal

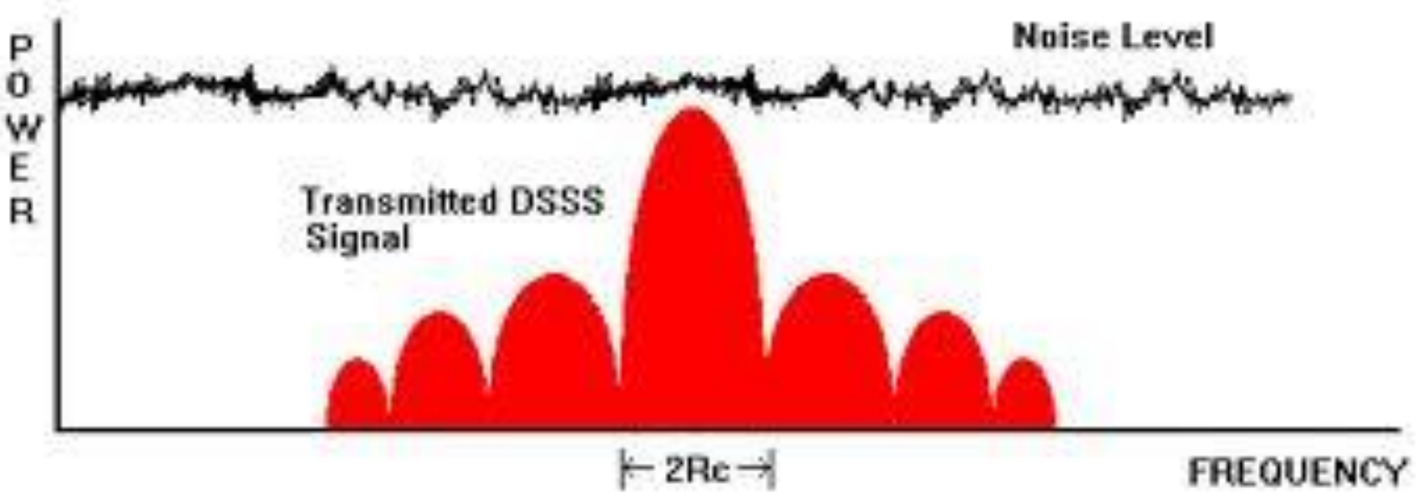
B A non-periodic signal

Submit

# 1.2 Signal Classification and Typical Signals

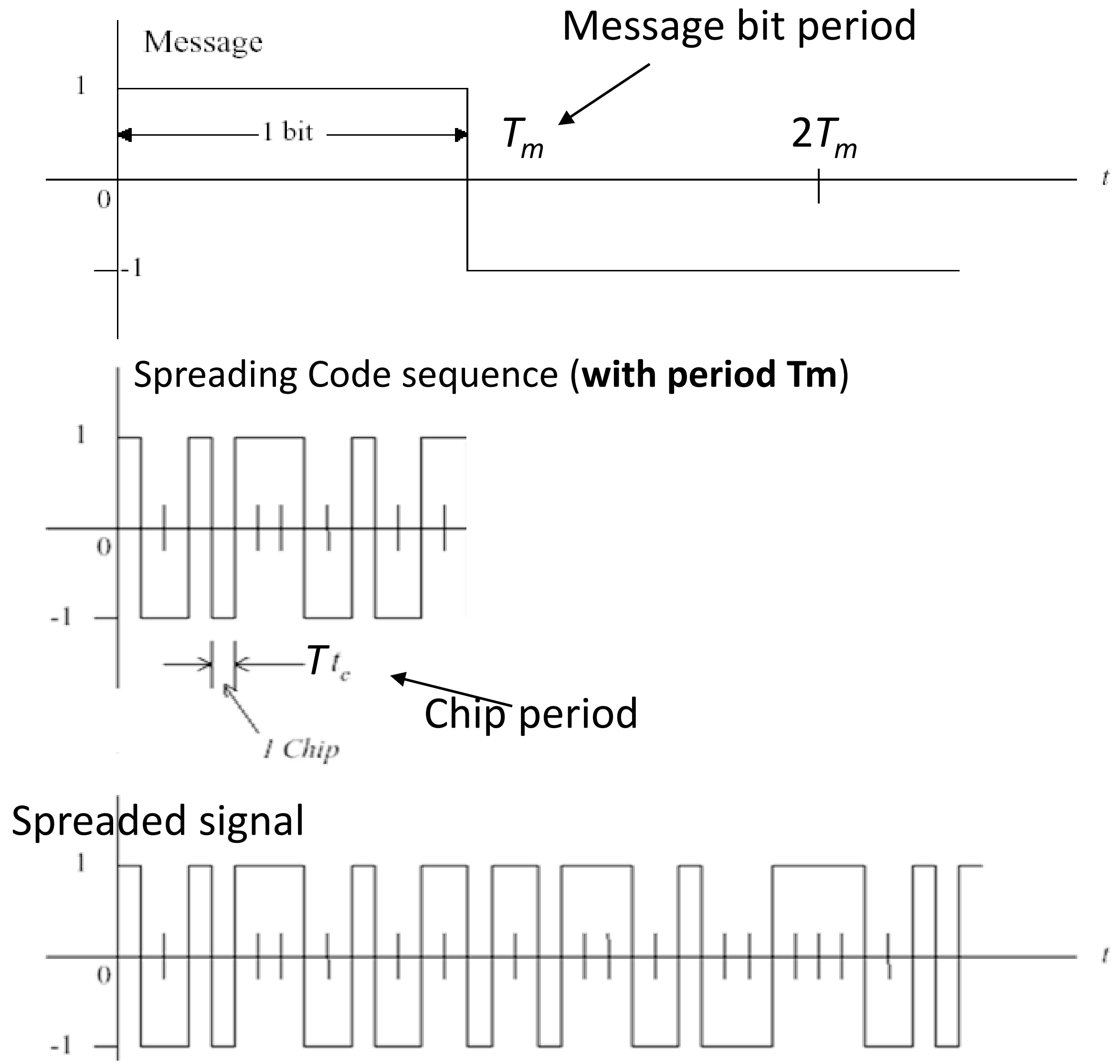
Direct Sequence (DS)-Spread Spectrum (SS): a modulation technique used in secure communication and 3G.



# 1.2 Signal Classification and Typical Signals

DS-SS

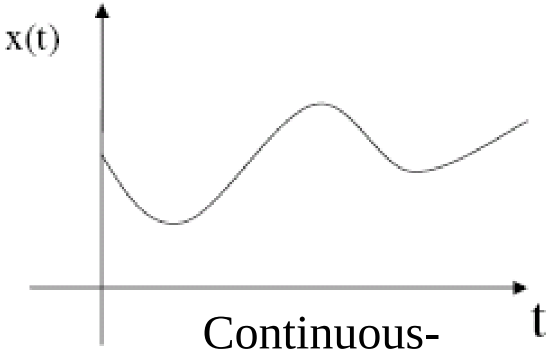


## 3. Continuous-Time Signals and Discrete-Time Signals

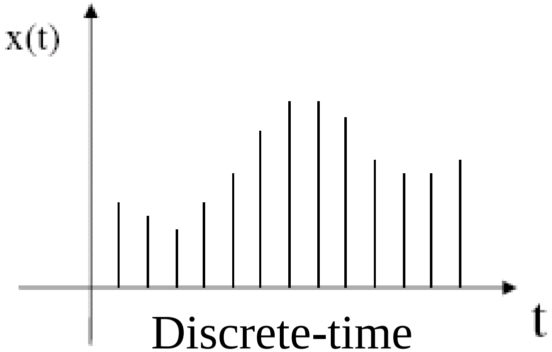
**Continuous-time (analog) signal**: Within the time interval under discussion, a definite function value can be given for any time value (except for several discontinuous points).

**Discrete-time signal**: The signal has values only at discrete time points and is undefined at other times. It is usually obtained via sampling of continuous-time signals.

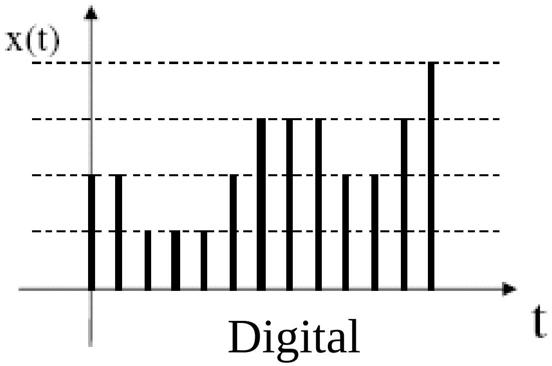
**Digital signal**: The signal is discrete in both time and amplitude. **Quantization** is the process of converting a continuous range of analog signal amplitudes (or highly precise discrete values) into a finite set of discrete levels.




Continuous-time signal




Discrete-time signal

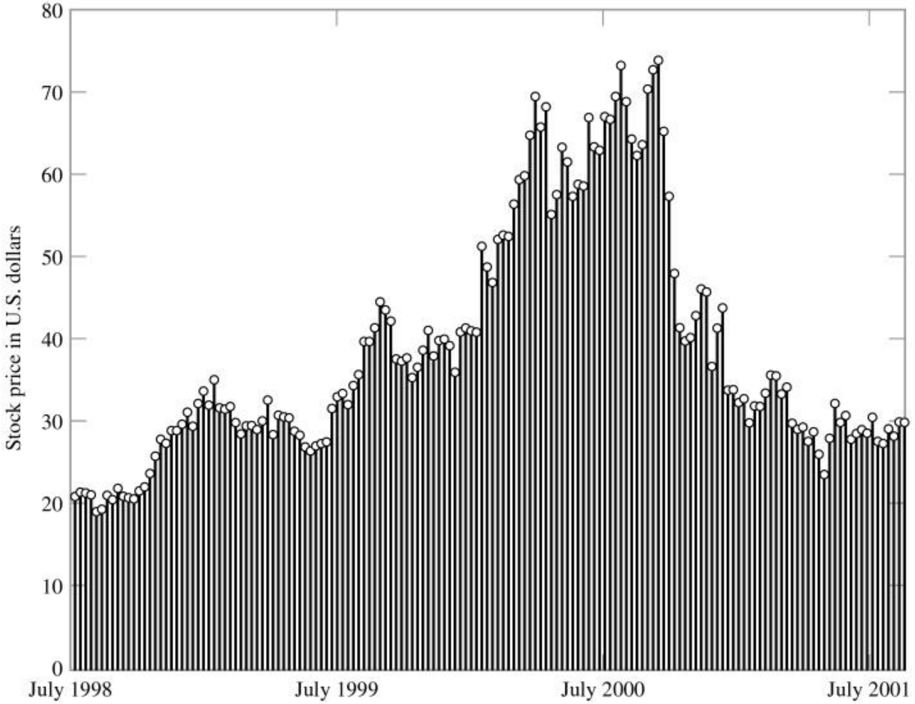



Digital signal

## 4. One-dimensional signals and multi-dimensional signals

A signal can be expressed as a function of one or more variables.

In this course, we mainly study **one-dimensional** signals where the variable is **time**.



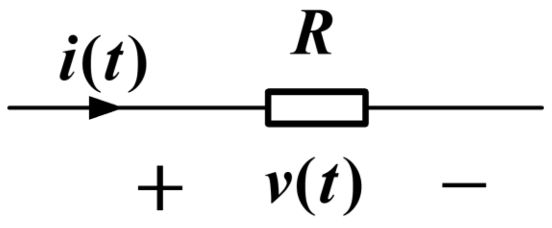

Stock variation signal-one dimensional signal

Picture—two-dimensional signal

## 5. Energy signals and power signals

**Let $i(t)$ be the current through resistor $R$, and $v(t)$ be the voltage across resistor $R$.**

**The instantaneous power is**




$$p(t) = i^2(t)R$$

**The energy consumed by R over a time period is**

$$E = \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} p(t)\,\mathrm{d}t = R\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} i^2(t)\,\mathrm{d}t \text{ 或 } \quad E = \frac{1}{R}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} u^2(t)\,\mathrm{d}t$$

**The average power is**

$$P = \frac{1}{T_0} R\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} i^2(t)\,\mathrm{d}t \text{ 或 } P = \frac{1}{T_0}\frac{1}{R}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} u^2(t)\,\mathrm{d}t$$

**Definition**: Generally, energy is always proportional to the square of a certain physical quantity.

Let $R$ = 1, then within the entire time domain, for a real signal $f(t)$,

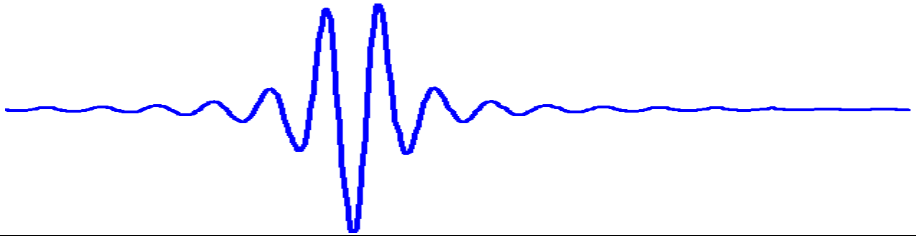
**Energy:** $$E = \lim_{T_0 \to \infty} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} f^2(t)\,\mathrm{d}t$$

**Average power:** $$P = \lim_{T_0 \to \infty} \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} f^2(t)\,\mathrm{d}t$$

**There are two possibilities:**

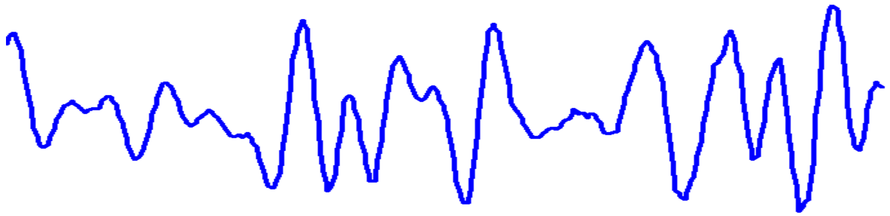
① $0 < E < \infty$ (with finite value),

$P = 0$ ⟵ **Energy signal**



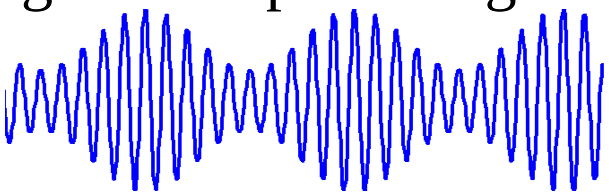
② $0 < P < \infty$ (with finite value),

$E = \infty$ ⟵ **Power signal**

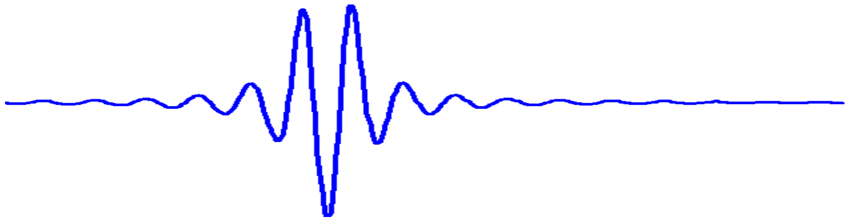


## General Rules

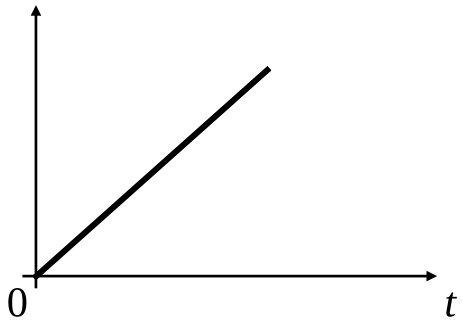
① Generally, periodic signals are power signals.



② Non-periodic signals with values within a limited interval are energy signals.



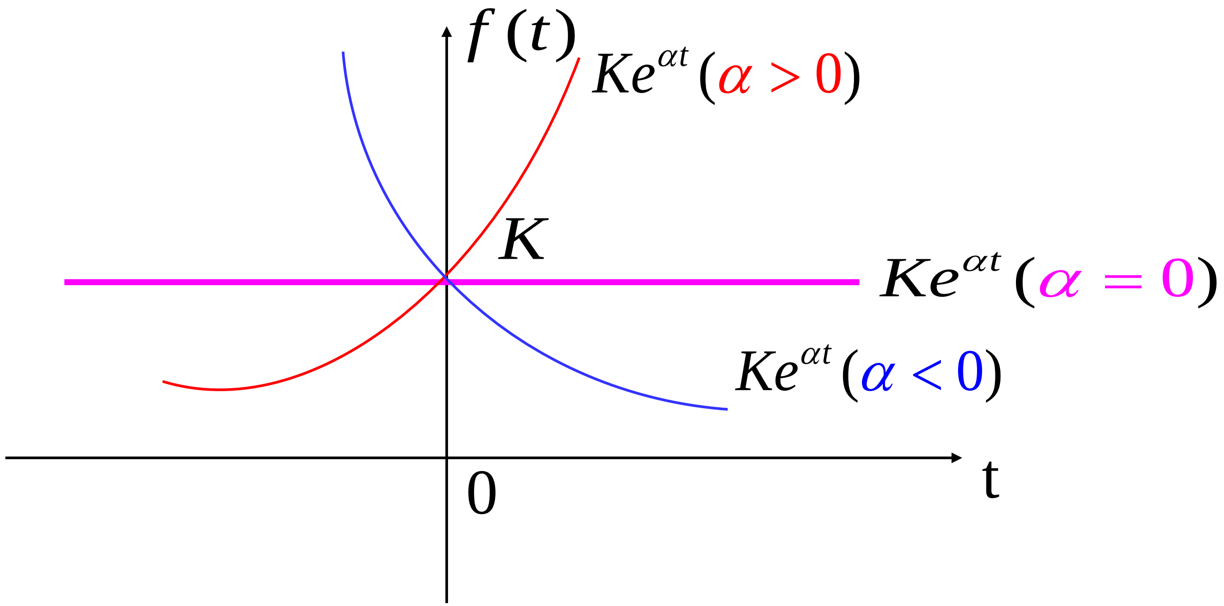
③ Some non-periodic signals are neither energy signals nor power signals.



## 1.2.2 Typical Signals

### 1. Exponential Signal

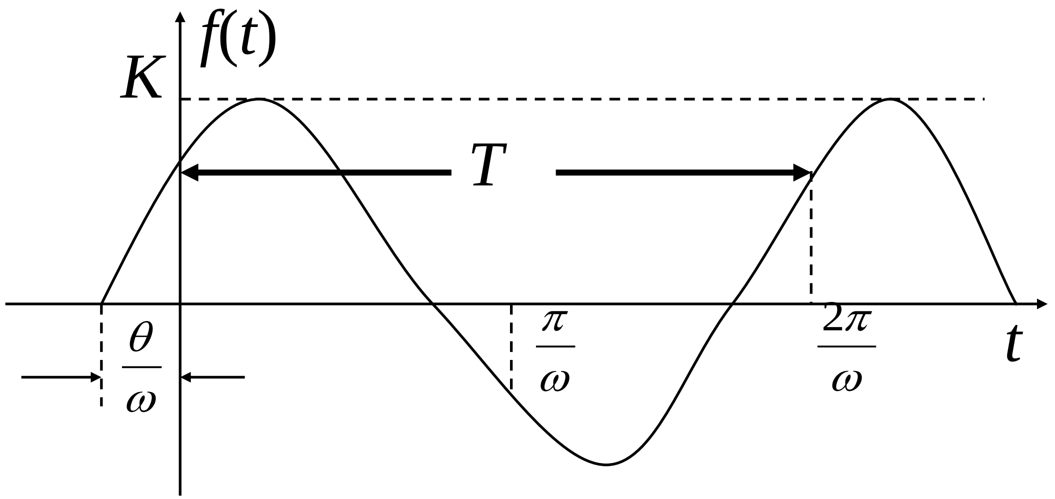
Expression: $f(t) = Ke^{\alpha t}$




The result of differentiating or integrating an exponential function with respect to time is still in exponential form.

## 2. Sinusoidal Signal

$$f(t) = K\sin(\omega t + \theta)$$




$$e^{j\omega t} = \cos\omega t + j\sin\omega t \qquad e^{-j\omega t} = \cos\omega t - j\sin\omega t$$

$$\sin\omega t = \frac{1}{2j}(e^{j\omega t} - e^{-j\omega t}) \qquad \cos\omega t = \frac{1}{2}(e^{j\omega t} + e^{-j\omega t})$$

## 3. Complex Exponential Signal

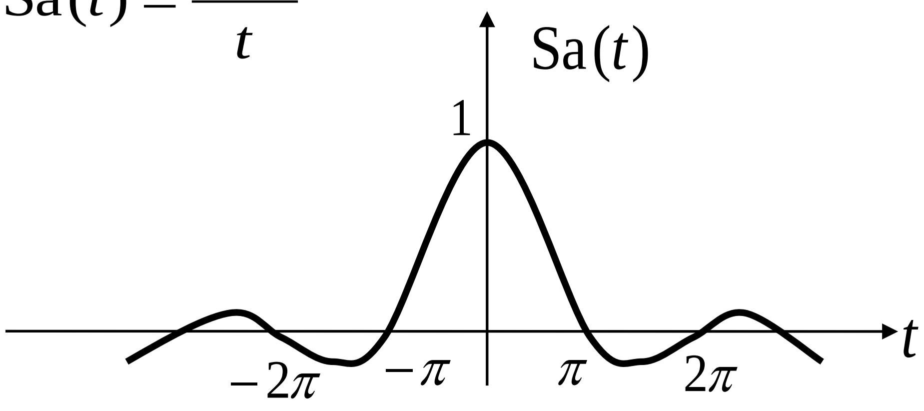
If the exponential factor of an exponential signal is a complex number, it is referred to as a complex exponential signal, expressed as:

$$f(t) = Ke^{st} = Ke^{(\sigma + j\omega)t} = Ke^{\sigma t}\cos\omega t + jKe^{\sigma t}\sin\omega t$$

## 4. Sampling function Sa(*t*) and Sinc(t)

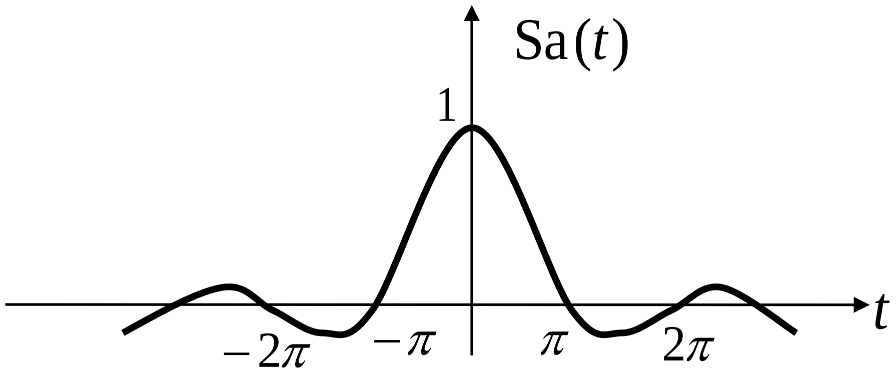
The sampling function is defined as:

$$\mathrm{Sa}(t) = \frac{\sin t}{t}$$




An alternative function is defined as:

$$\mathrm{Sinc}(t) = \frac{\sin(\pi t)}{\pi t}$$




Properties of $\mathrm{Sa}(t)$

（1） $\mathrm{Sa}(t)$ is an even function， and its amplitude gradually decays in both the positive and negative directions of *t*.

（2） $\int_0^{\infty} \mathrm{Sa}(t)dt = \frac{\pi}{2}$ $\qquad$ $\int_{-\infty}^{\infty} \mathrm{Sa}(t)dt = \pi$

## 1.3.1 Addition of Signals

The sum (or difference) of two signals is still a signal, and its value at any time is equal to the sum (or difference) of the values of the two signals at that time, i.e.,

$$f(t) = f_1(t) + f_2(t) \quad \text{or} \quad f(t) = f_1(t) - f_2(t)$$

## 1.3.2 Multiplication and Scaling of Signals

The product of two signals is still a signal, and its value at any time is equal to the product of the values of the two signals at that time, i.e.,

$$f(t) = f_1(t) f_2(t)$$

The scaling operation of a signal refers to multiplying a signal by a real constant K, which means multiplying the value of the original signal at each time by *K*, i.e.,

$$f(t) = Kf(t)$$

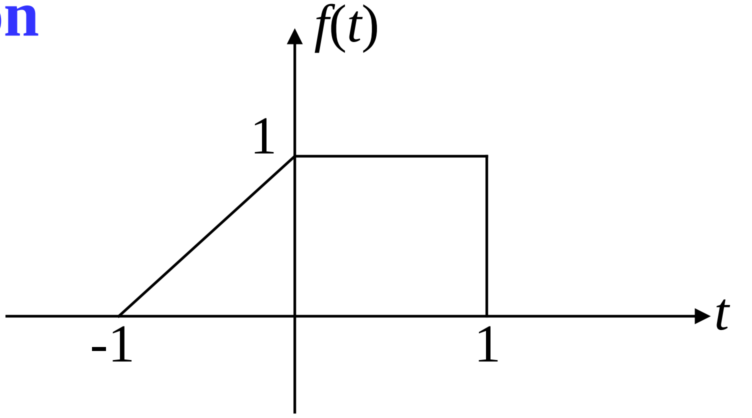
# 1.3 Signal Operations

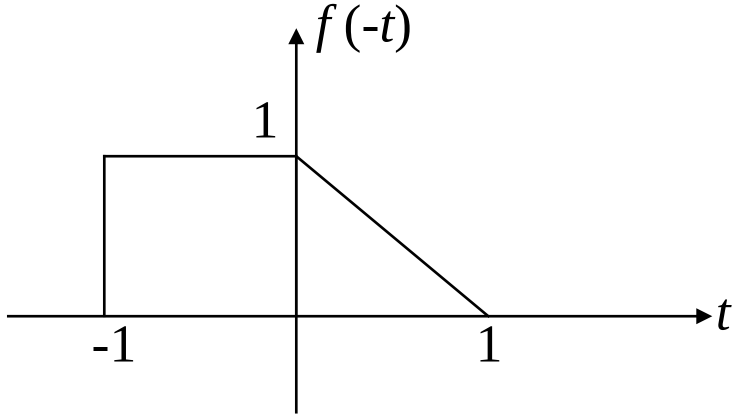
## 1.3.3 Time Operations of Signals-Reverse, Shifting and Scaling

### 1. Reverse Operation

$$f(t) \rightarrow f(-t)$$

**Reverse against the *t* = 0 axis.**



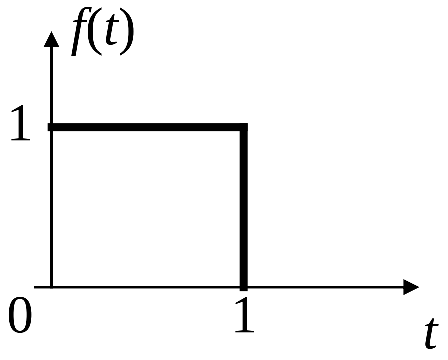



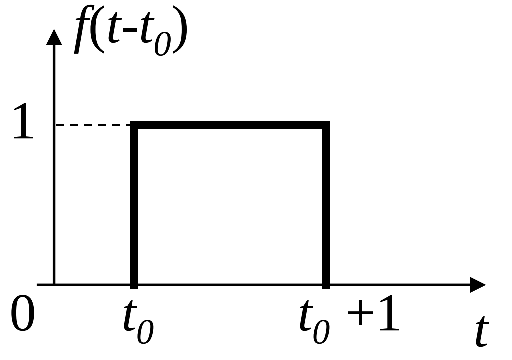

### 2. Time Shifting Operation

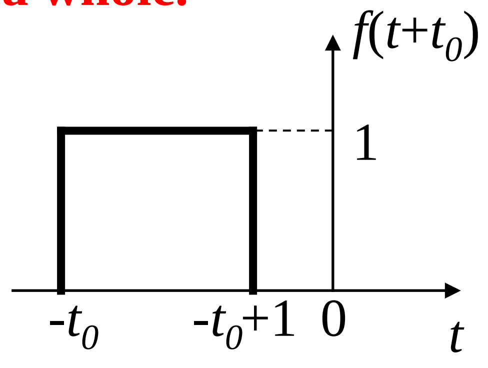
$$f(t) \rightarrow f(t-t_0)$$

**$t_0$>0时，$f(t)$ shifts to the right by $t_0$ as a whole.**

**$t_0$<0时，$f(t)$ shifts to the left by $t_0$ as a whole.**



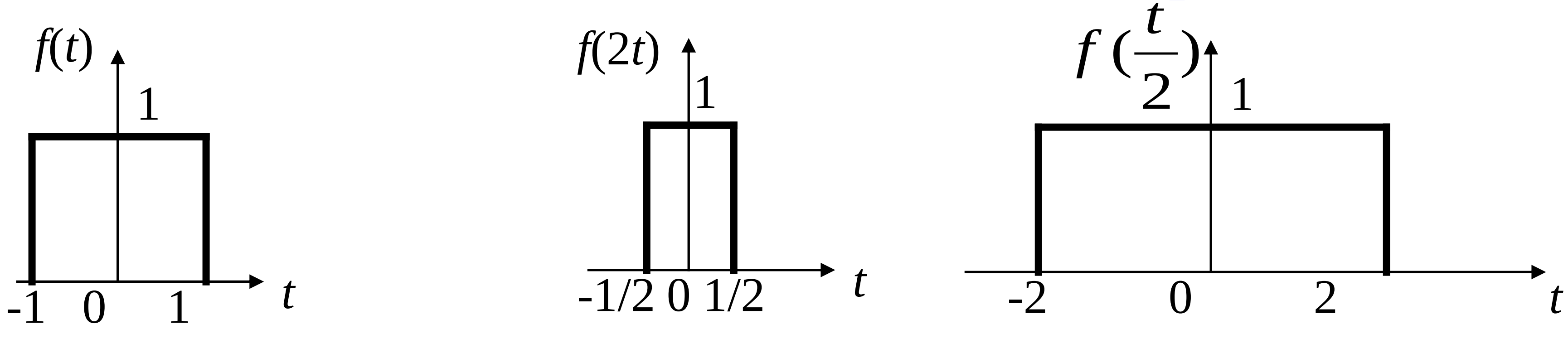



# 1.3 Signal Operations

## 3. Scaling Operation

$$f(t) \to f(2t) \qquad\qquad f(t) \to f\left(\frac{t}{2}\right)$$

**Compression** **Expansion**




$$f(t) \to f(at)$$

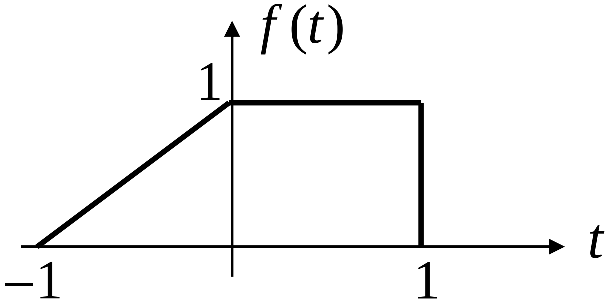
**Compression**: If $|a| > 1$, the waveform is compressed along the time axis.

**Expansion**: If $0 < |a| < 1$, the waveform is expanded along the time axis.

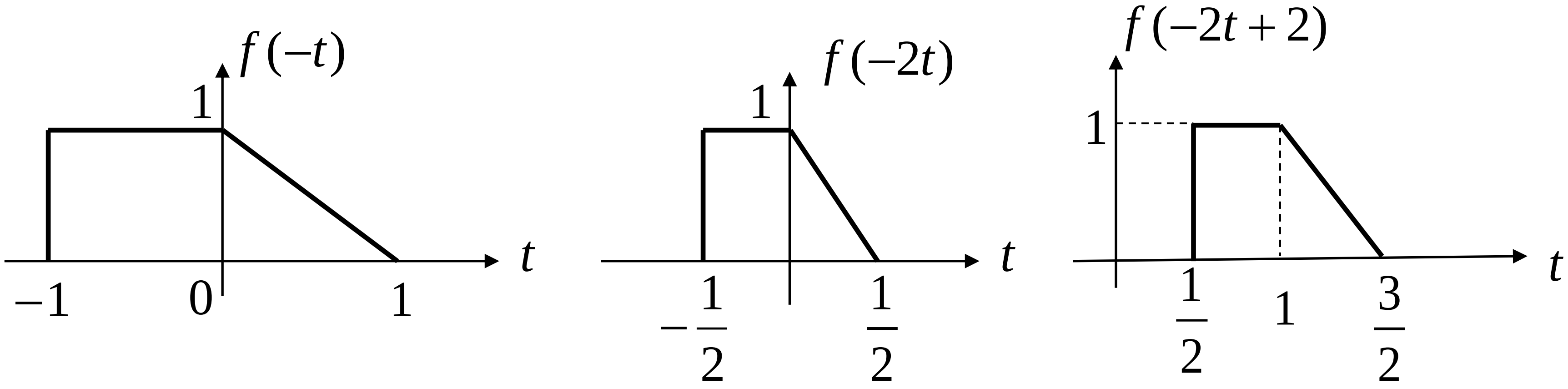
# 1.3 Signal Operations

Please give one real-life example each for signal reverse, time shifting and compression.

**Example 1-1**：**The signal is shown in the figure below, find f(-2t+2), and sketch the waveform.**




$$f(t) \xrightarrow{\text{reverse}} f(-t) \xrightarrow{\text{scaling}} f(-2t) \xrightarrow{\text{shift}} f(-2t+2) = f[-2(t-1)]$$

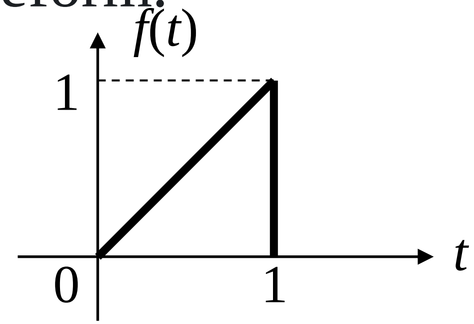


## 1.3.4 Differentiation and Integration Operations of Signals

### 1. Differentiation Operation

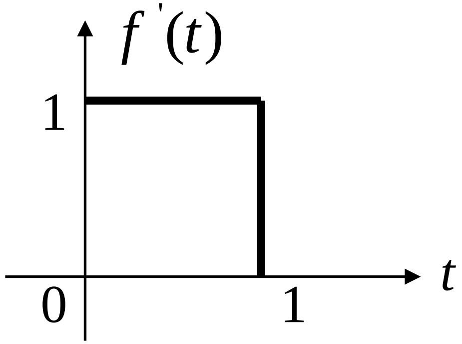
The derivative of signal $f(t)$ is still a signal, represented by $f'(t)$ ,which represents the rate of change of the signal with time. The differentiation operation highlights the changing part of the signal.

**Example 1-2**: Find the derivative $f'(t)$ of signal $f(t)$ shown in the figure below and sketch its waveform.




**Solution:** $f(t) = t \quad (0<t\leq 1)$

$f'(t) = 1 \quad (0<t\leq 1)$

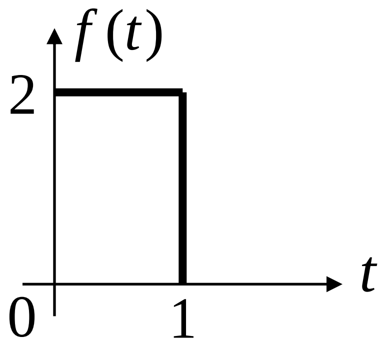


## 2. Integration Operation

•The integral of signal $f(t)$, denoted by $\int_{-\infty}^{t} f(\tau)d\tau$ or $f^{(-1)}(t)$, is still a signal. Its value at any time is equal to the area enclosed by f(t) and the time axis within the interval from -∞ to t.

**•The integration operation smooths the abrupt parts of the signal and can reduce the impact of glitches (noise).**

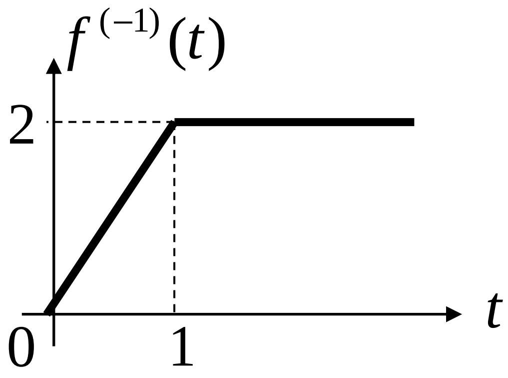
**Example 1-3:** Select the waveform of the integral $f^{(-1)}(t)=\int_{-\infty}^{t} f(\tau)d\tau$ of signal $f(t)$ shown in the figure.




**Solution :** 1）If $t<0$时， $f^{-1}(t)=0$

2）If $0\le t\le 1$ 时， $f^{(-1)}(t)=\int_{0}^{t}2d\tau=2t$

3）If $t>1$ 时， $f^{-1}(t)=\int_{0}^{1}2d\tau=2$

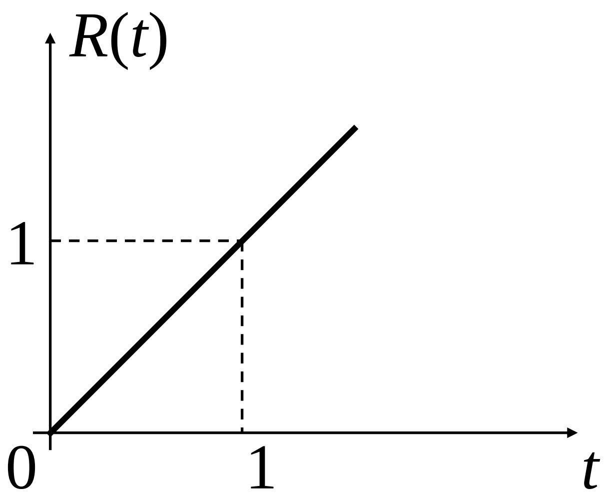


In the analysis of signals and systems, we often encounter functions with discontinuities in themselves or their derivatives/integrals. Such functions are collectively called singular functions or singular signals.

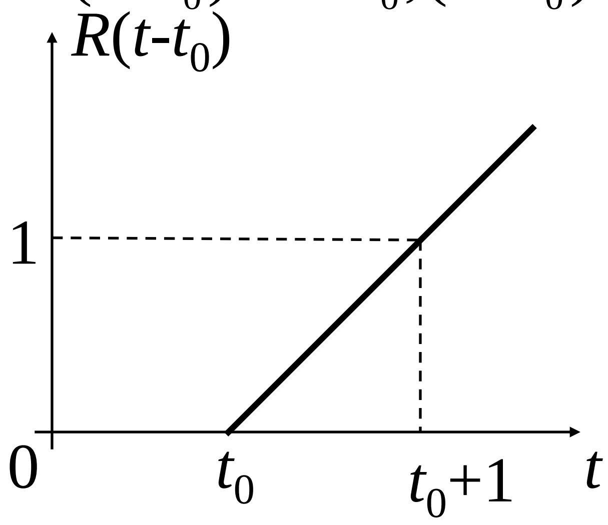
## 1.4.1 Unit Ramp Signal

A unit ramp signal refers to a signal that increases proportionally with a unit slope, starting from time 0.

$$R(t) = t, (t \geq 0)$$

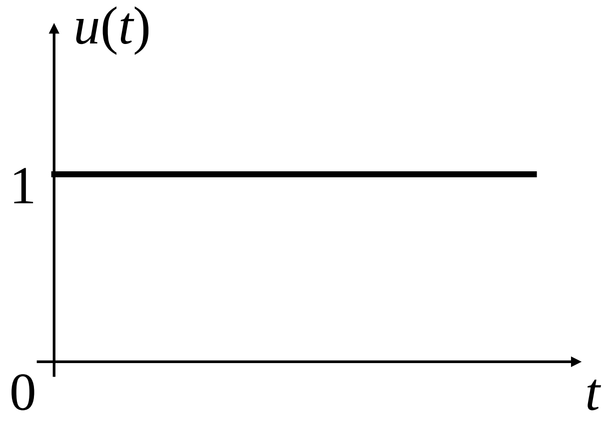



$$R(t-t_0) = t - t_0, (t \geq t_0)$$

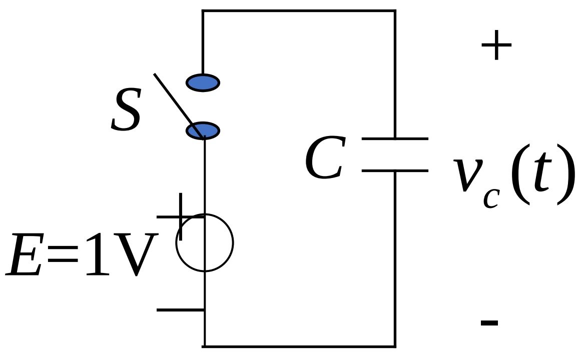


## 1.4.2 Unit Step Signal

$$u(t) = \begin{cases} 0, (t < 0) \\ 1, (t > 0) \end{cases}$$




What is the physical meaning of the unit step signal in circuits?

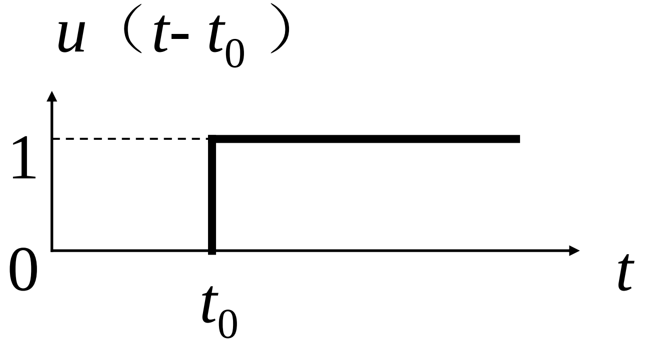



**Example 1-4:** Assume that S, E, and C in the figure are all ideal components (internal resistance = 0). Find the voltage across capacitor C when S is closed at t = 0.

**Solution:** Since S, E, and C are all ideal components, there is no internal resistance in the circuit. After S is closed, the voltage across C will jump, forming a step voltage. That is:

$$v_c(t) = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases} = u(t)$$

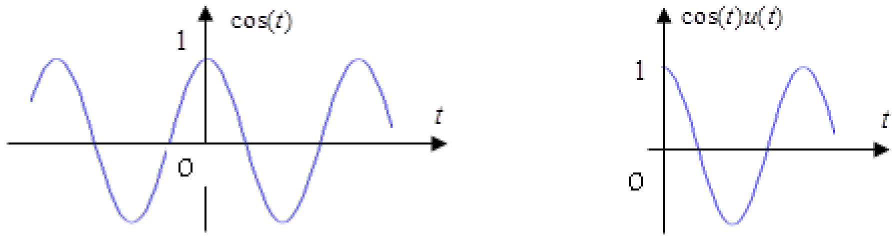
If switch S is closed at $t = t_0$, the voltage across the capacitor is $u(t - t_0)$. The waveform is shown below.



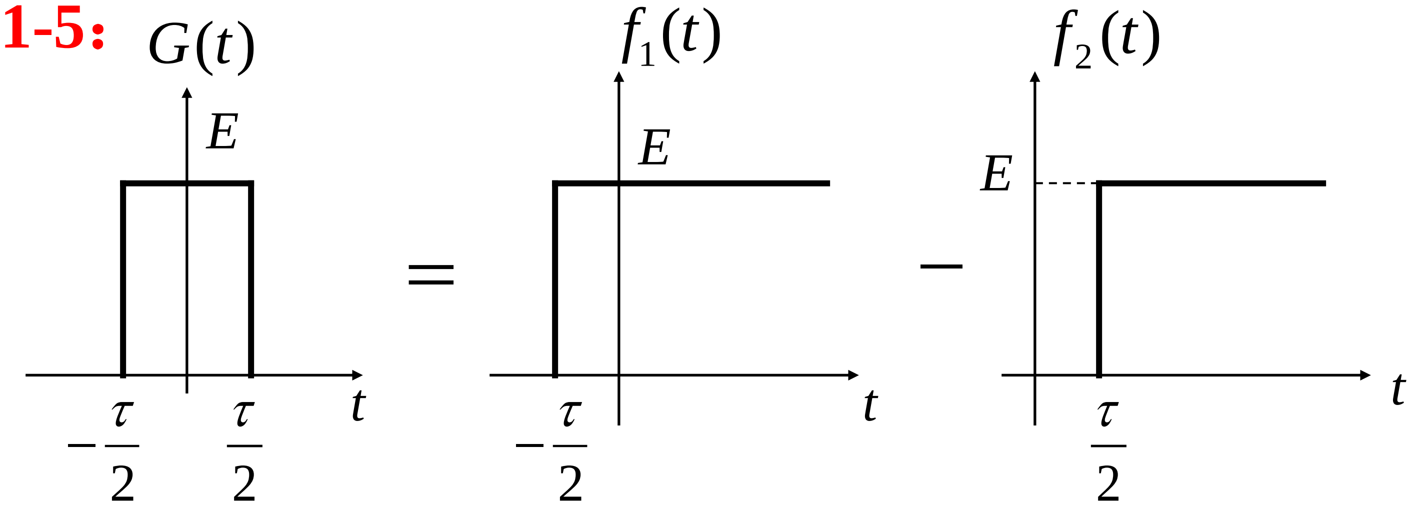
**Properties of *u*(*t*):** Unilateral characteristic, i.e.,

$$f(t)u(t)=\begin{cases}0 & t<0\\ f(t) & t>0\end{cases}$$

Some pulse signals can be expressed using step signals.



**Example 1-5:**

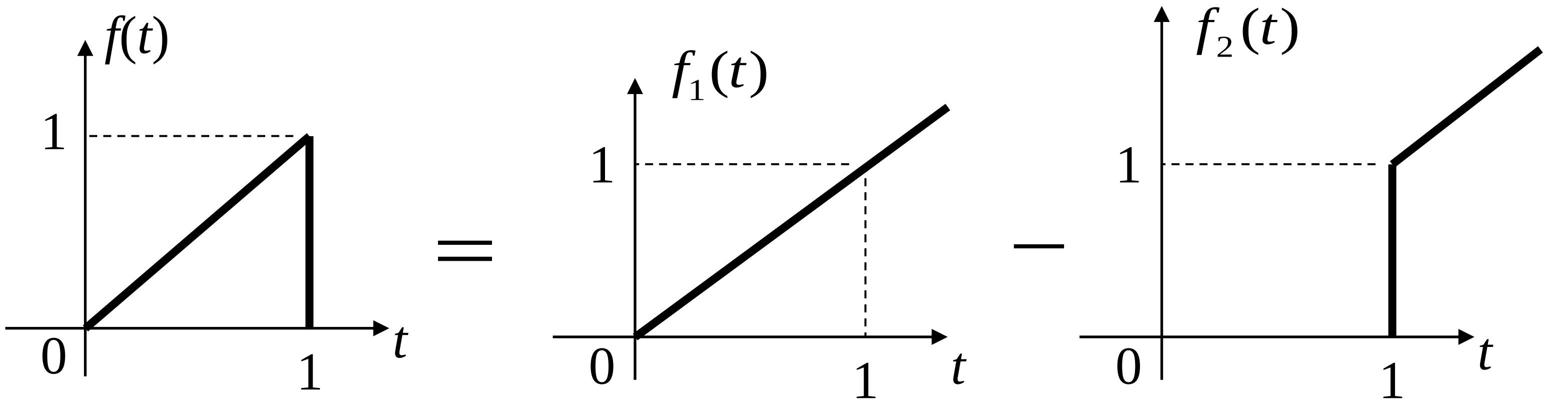



Since $f_1(t) = Eu(t+\frac{\tau}{2}),\qquad f_2(t) = Eu(t-\frac{\tau}{2}),$

the rectangular pulse $G(t)$ can be expressed as:

$$G(t) = f_1(t) - f_2(t) = E[u(t+\frac{\tau}{2}) - u(t-\frac{\tau}{2})]$$

**Example 1-6:**

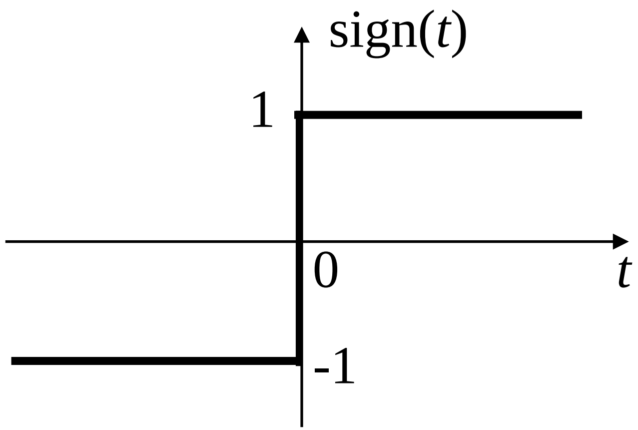



$$f(t)=t[u(t)-u(t-1)]$$

# 单选题 1分

设置

**Example 1-7：** Which of the following is the correct expression for the "signum function" (sign(t)) using step signals?




A $1-2u(t)$

B $2u(t)-1$

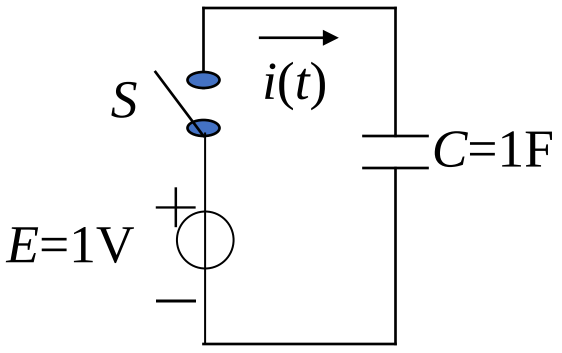
C $2u(t)-2$

D $2-2u(t)$

提交

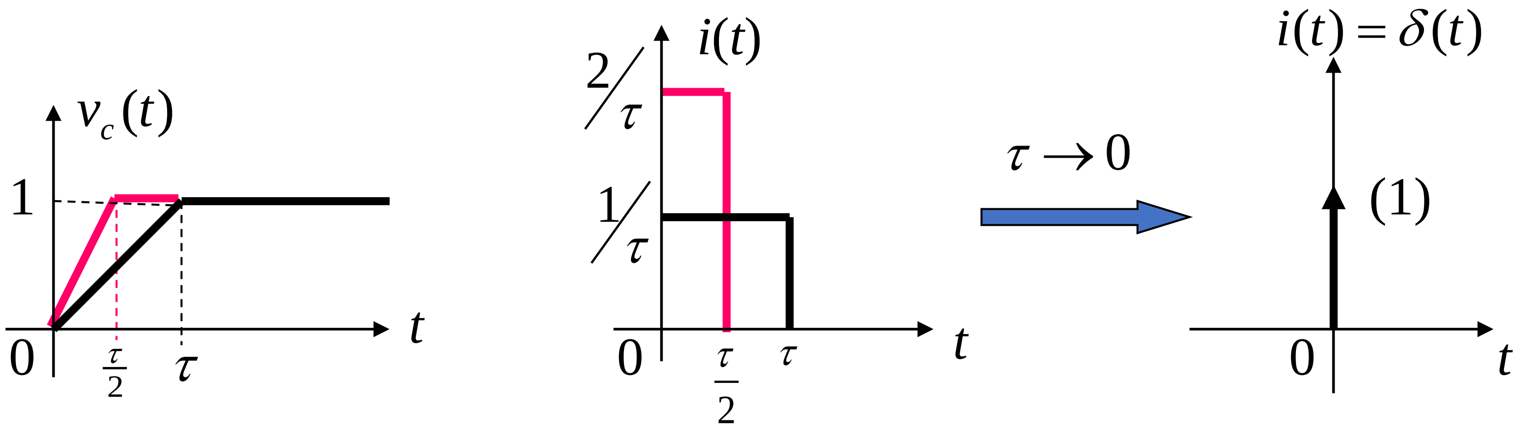
## 1.4.3 Unit Impulse Signal (Dirac Delta Function)

Physical meaning of the unit impulse signal (Dirac delta function, $\delta(t)$) with the following example:




Assume that S, E, and C in the figure are all ideal components (internal resistance = 0). Find current $i(t)$ when S is closed at t = 0.

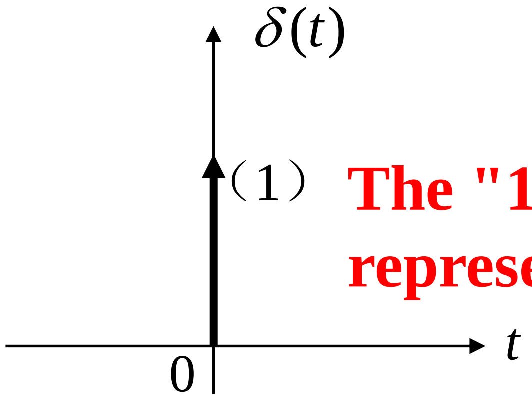
$$i(t) = C\frac{dv_C(t)}{dt}$$



## 1. Definition of $\delta(t)$

### 1）By Expression

$$\begin{cases} \delta(t) = 0 \quad (t \neq 0) \\ \int_{-\infty}^{\infty} \delta(t)dt = 1 \end{cases}$$

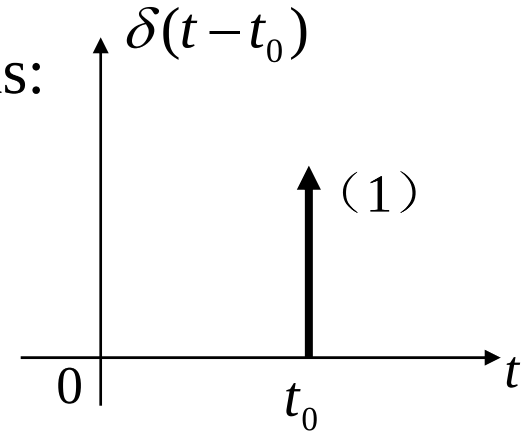



**The "1" in the bracket represents the strength.**

This definition method was proposed by Dirac.

Similarly, $\delta(t-t_0)$ can be defined as:

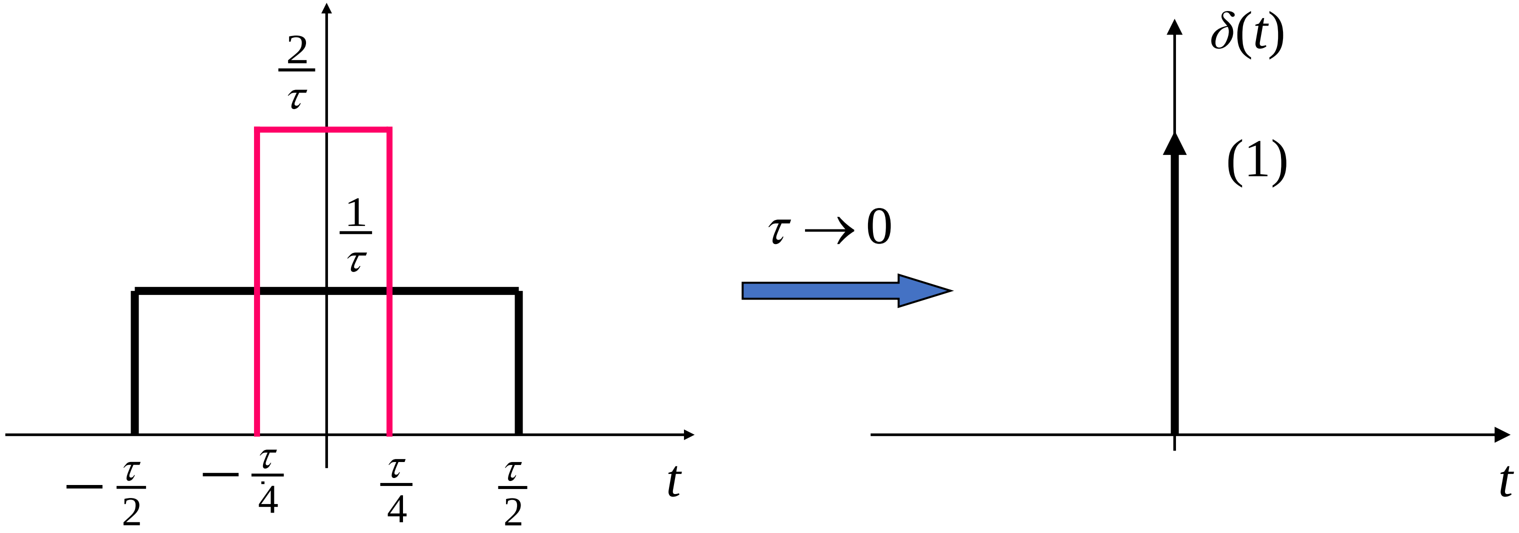
$$\begin{cases} \delta(t-t_0) = 0 \quad (t \neq t_0) \\ \int_{-\infty}^{\infty} \delta(t-t_0)dt = 1 \end{cases}$$



## 2) By Limit

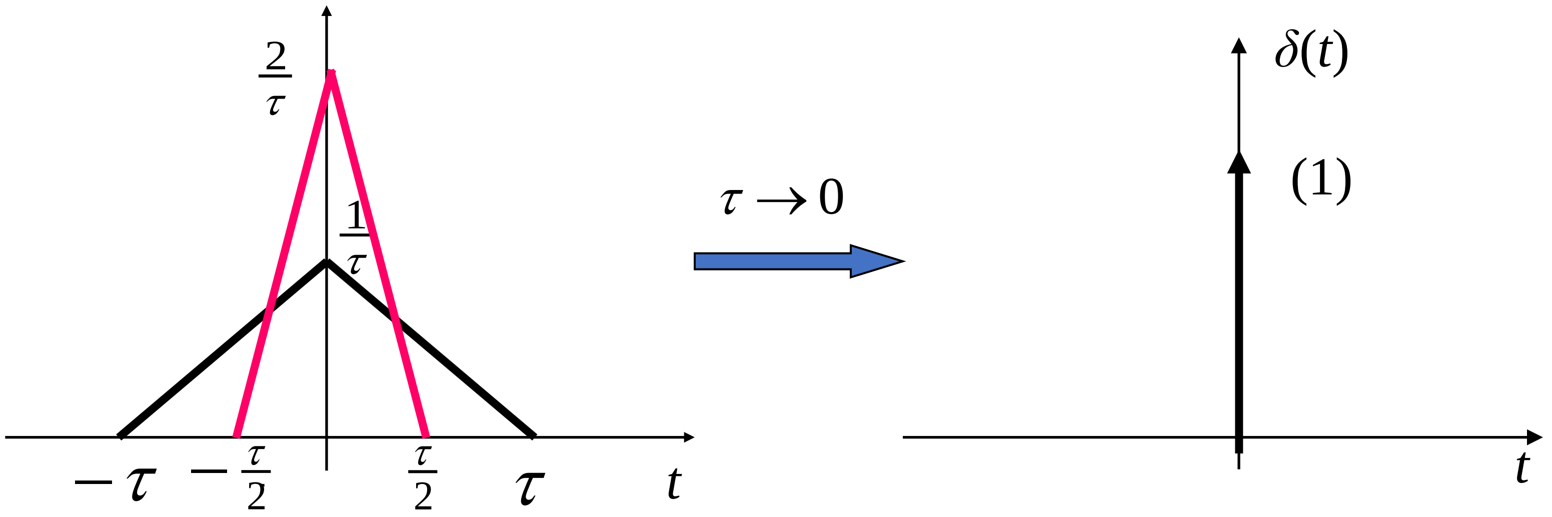
We can define δ(t) by taking the limit of various regular function sequences.

For example, (a) definition by the limit of rectangular pulses




$$\delta(t) = \lim_{\tau \to 0} \frac{1}{\tau}[u(t+\frac{\tau}{2}) - u(t-\frac{\tau}{2})]$$

（b）Definition by the limit of triangular pulses




$$\delta(t) = \lim_{\tau \to 0} \left\{ \frac{1}{\tau}(1 - \frac{|t|}{\tau})[u(t+\tau) - u(t-\tau)] \right\}$$

## 2. Properties of the Unit Impulse Signal

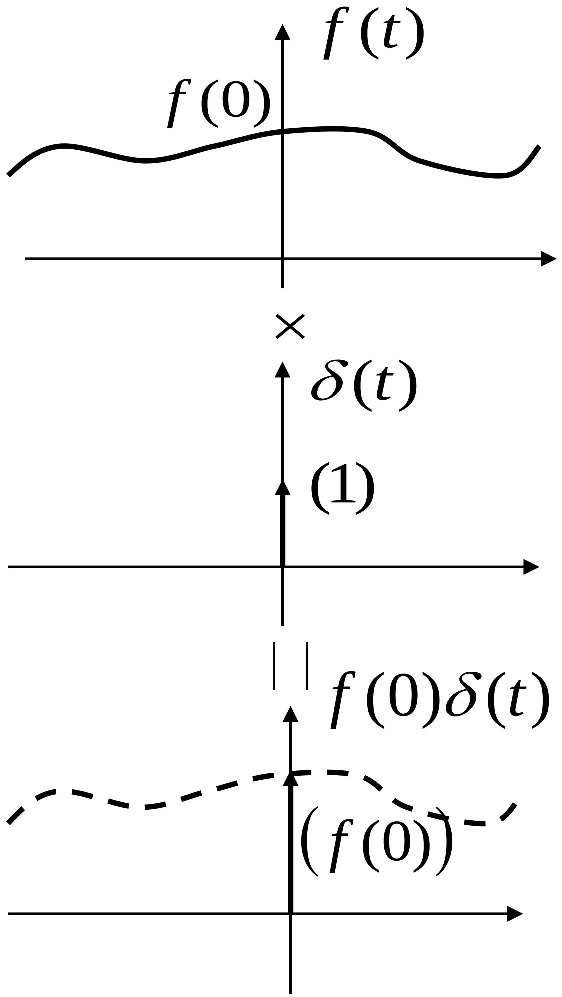
### 1）Sampling

$$f(t)\delta(t) = f(0)\delta(t) \qquad (1)$$

$$\int_{-\infty}^{\infty} \delta(t) f(t) dt = f(0)\int_{-\infty}^{\infty} \delta(t) dt = f(0) \qquad (2)$$

$$f(t)\delta(t-t_0) = f(t_0)\delta(t-t_0) \qquad (3)$$

$$\int_{-\infty}^{\infty} \delta(t-t_0) f(t) dt = f(t_0) \qquad (4)$$



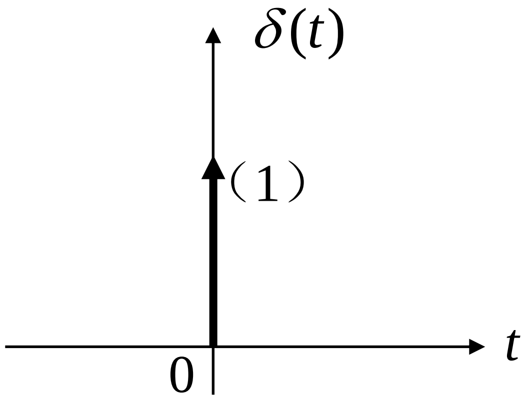

Combining formulas (2) and (4), we can draw the following conclusion:

**The impulse function can extract (sample) the function value at the position where the impulse is located.**

**2)** $\delta(t)$ is an even function, i.e., $\delta(t)=\delta(-t)$

**3)** $\int_{-\infty}^{t}\delta(\tau)d\tau = \begin{cases}0 & t<0\\ 1 & t>0\end{cases} = u(t)$

$$\int_{-\infty}^{t}\delta(\tau-t_0)d\tau = u(t-t_0)$$



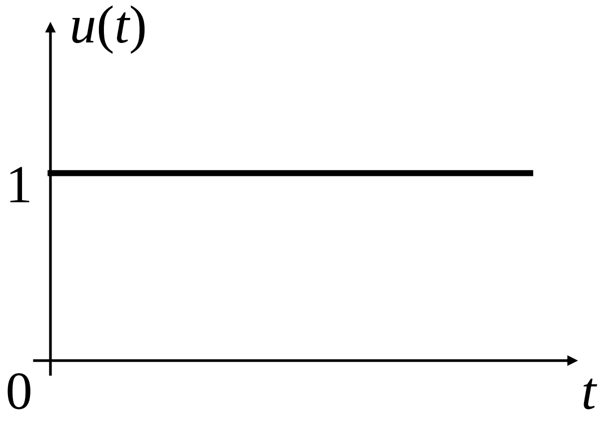

Relational ship between $u(t)$ and $\delta(t)$

$$\int_{-\infty}^{t}\delta(\tau)d\tau = u(t)$$

$$\frac{d}{dt}u(t)=\delta(t)$$

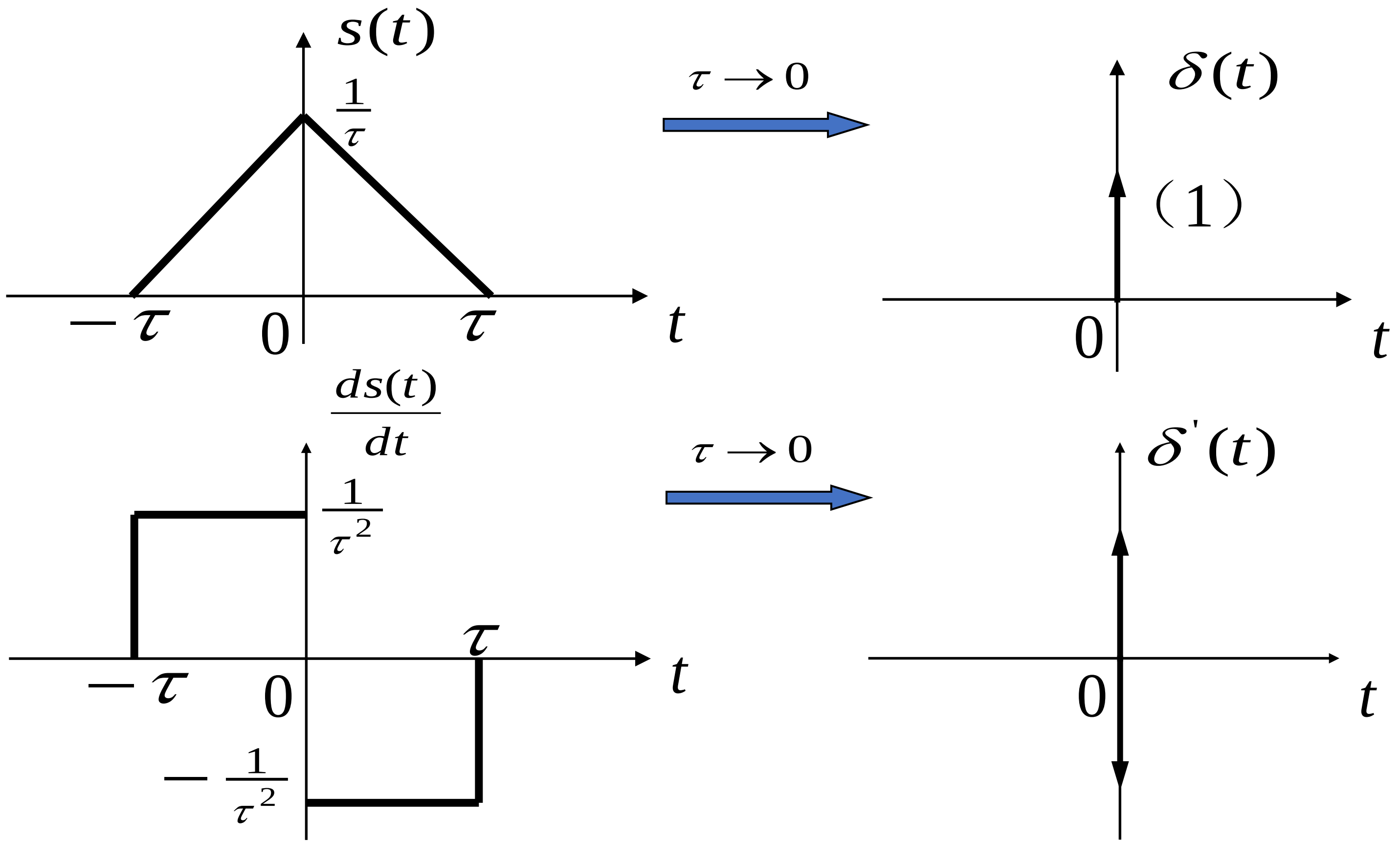
$$\int_{-\infty}^{t}\delta(\tau-t_0)d\tau = u(t-t_0)$$

$$\frac{d}{dt}u(t-t_0)=\delta(t-t_0)$$



## 1.4.4 Impulse Derivative (Doublet)

The derivative of the impulse function (second derivative of the step function) presents a pair of impulses with positive and negative polarities, which is called the impulse derivative (doublet), denoted by δ’(t).



## Properties of the Impulse Derivative

1） The impulse derivative is an odd function, i.e., $\delta^{'}(-t) = -\delta^{'}(t)$

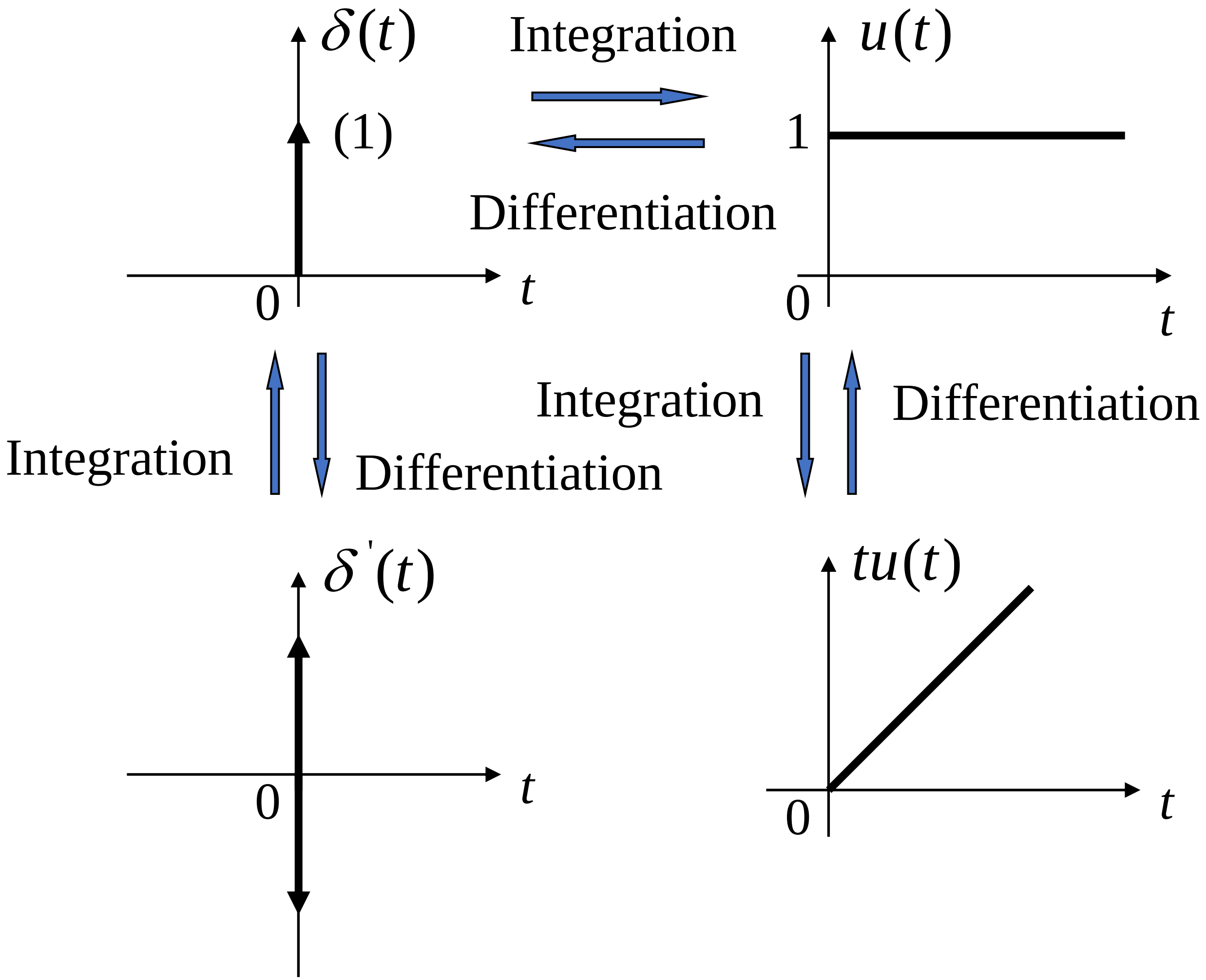
2） $$\int_{-\infty}^{\infty} \delta^{'}(t) f(t) dt = -f^{'}(0)$$

$$\int_{-\infty}^{\infty} \delta^{'}(t-t_0) f(t) dt = -f^{'}(t_0)$$

3） $$\int_{-\infty}^{\infty} \delta^{'}(t) dt = 0$$

# 1.4 Singular Signals

Relationship between $tu(t)$, $u(t)$, $\delta(t)$, and $\delta'(t)$:



# Homework

- Homework: 1.50, 1.51, and 1.57(c)(f)(g) in the textbook.
- Textbook: Simon Haykin and Barry Van Veen, Signals and Systems (Second Edition), John Wiley & Sons, 2002, 9780471164746.
- You do not need to submit the homework, but you are highly recommended to do the exercises. Refer to the solutions after you finish the homework.

## Review of Last Lecture

## 1.1 Introduction

Message: A symbol to be transmitted in a way agreed in advance by both the sender and receiver, such as language, text, images, data, etc.

Information: Unknown content obtained from the received message.

Signal: A signal is a **variation** of a measurable quantity that carries **information** about the behaviour of a system. Examples: human voice, room temperature controlled by a thermostat system, position, speed, and acceleration of an aircraft, force measured with force sensors in robotic systems, digital photographs, music recording etc.

Electrical signal: The current or voltage that changes corresponding to the message (language, text, image, data), or the charge on the capacitor, the magnetic flux in the inductor, etc.

## 1.2.1 Classification of Signals

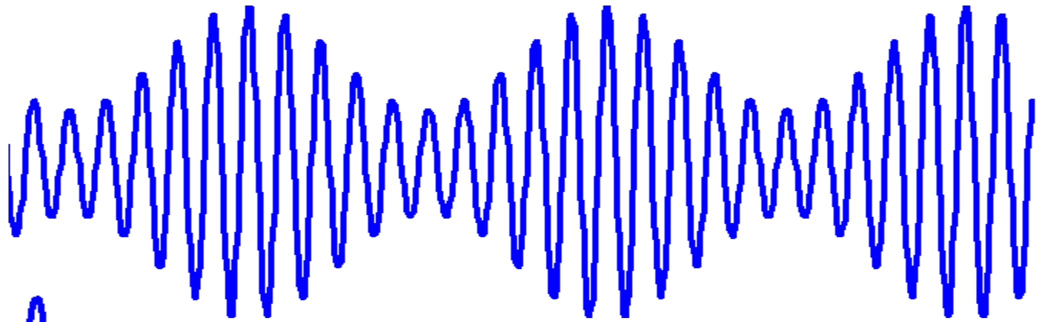
### 1. Deterministic Signals and Random Signals

Deterministic signal - for a certain moment, the signal has a definite value corresponding to it.

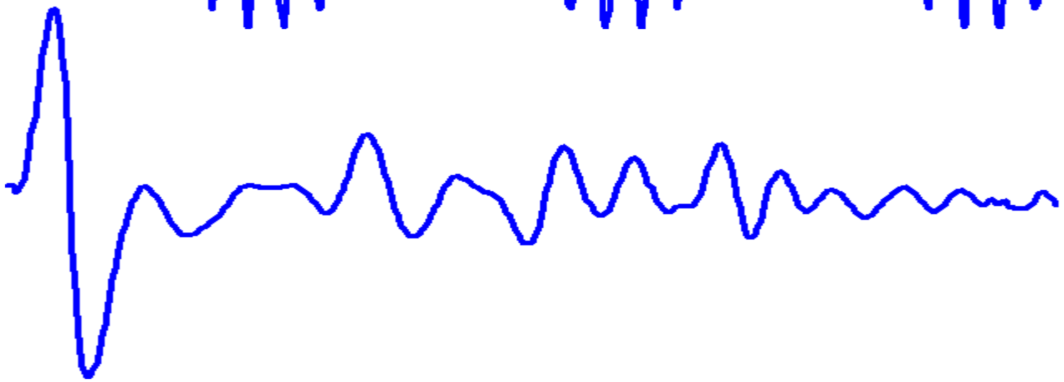
Random signals - unpredictable signals such as noise.

Deterministic signals

$S(t) = (2 + \cos\omega t)\cos(10\omega t)$



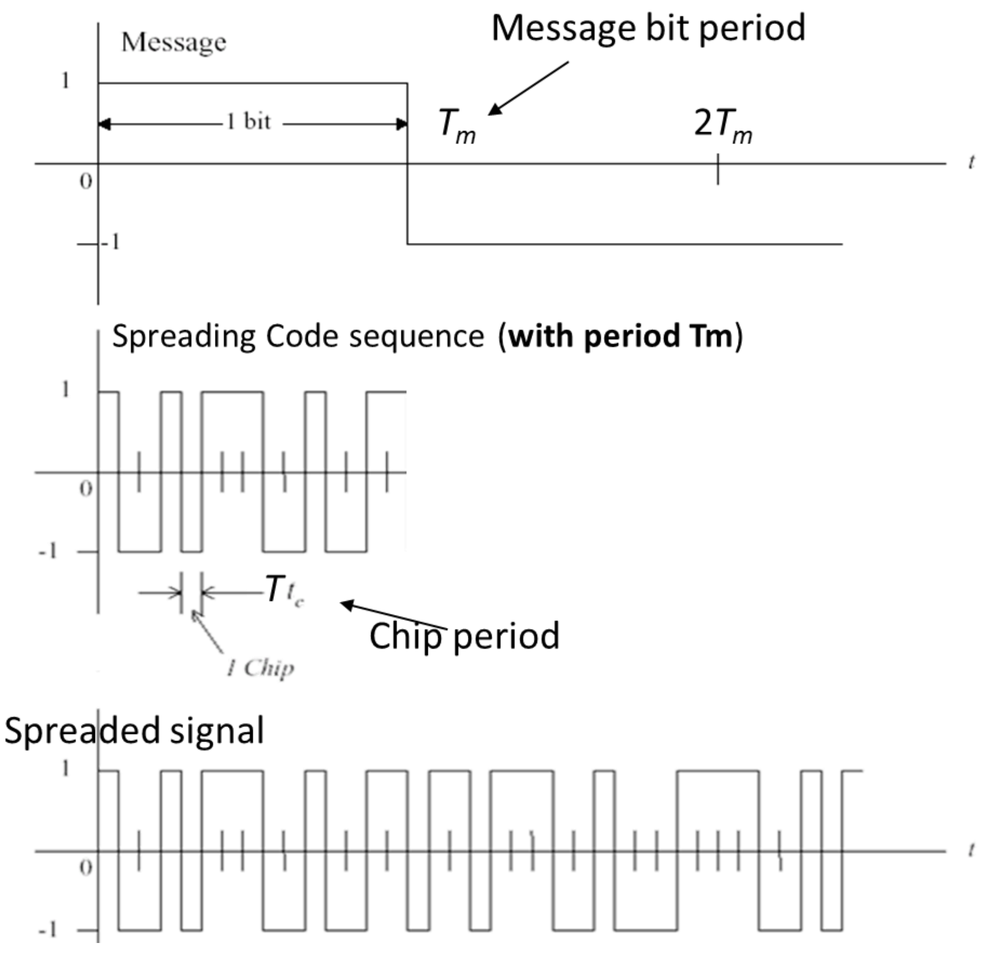
Random signal



## 2. Periodic Signals and Non-Periodic Signals

Periodic signal: It starts and starts at a certain time interval, and it is a signal without beginning or end. A signal with a basic period of $T$- $f(t)=f(t+T)$ for all $t$.

Non-periodic signal: A signal that does not meet the characteristics of the cycle in time.



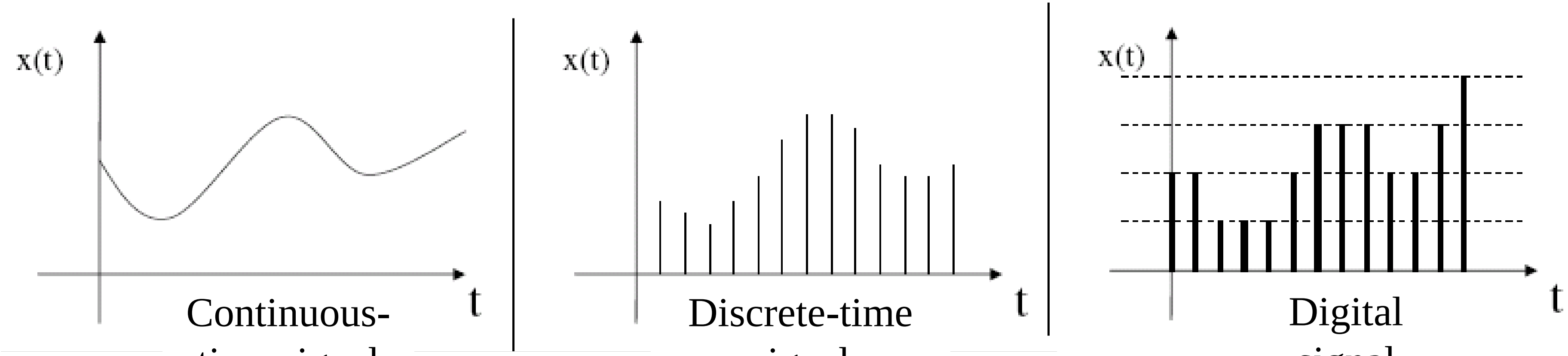

The spreading code of direct-sequence spread spectrum is a periodic signal.

## 3. Continuous-Time Signals and Discrete-Time Signals

**Continuous-time** (**analogue**) **signal**: Within the time interval under discussion, a definite function value can be given for any time value (except for several discontinuous points).

**Discrete-time signal**: The signal has values only at discrete time points and is undefined at other times. It is usually obtained via sampling of continuous-time signals.

**Digital signal**: The signal is discrete in both time and amplitude. **Quantization** is the process of converting a continuous range of analog signal amplitudes (or highly precise discrete values) into a finite set of discrete levels.




Continuous-time signal | Discrete-time signal | Digital signal

## 单选题 1分

设置

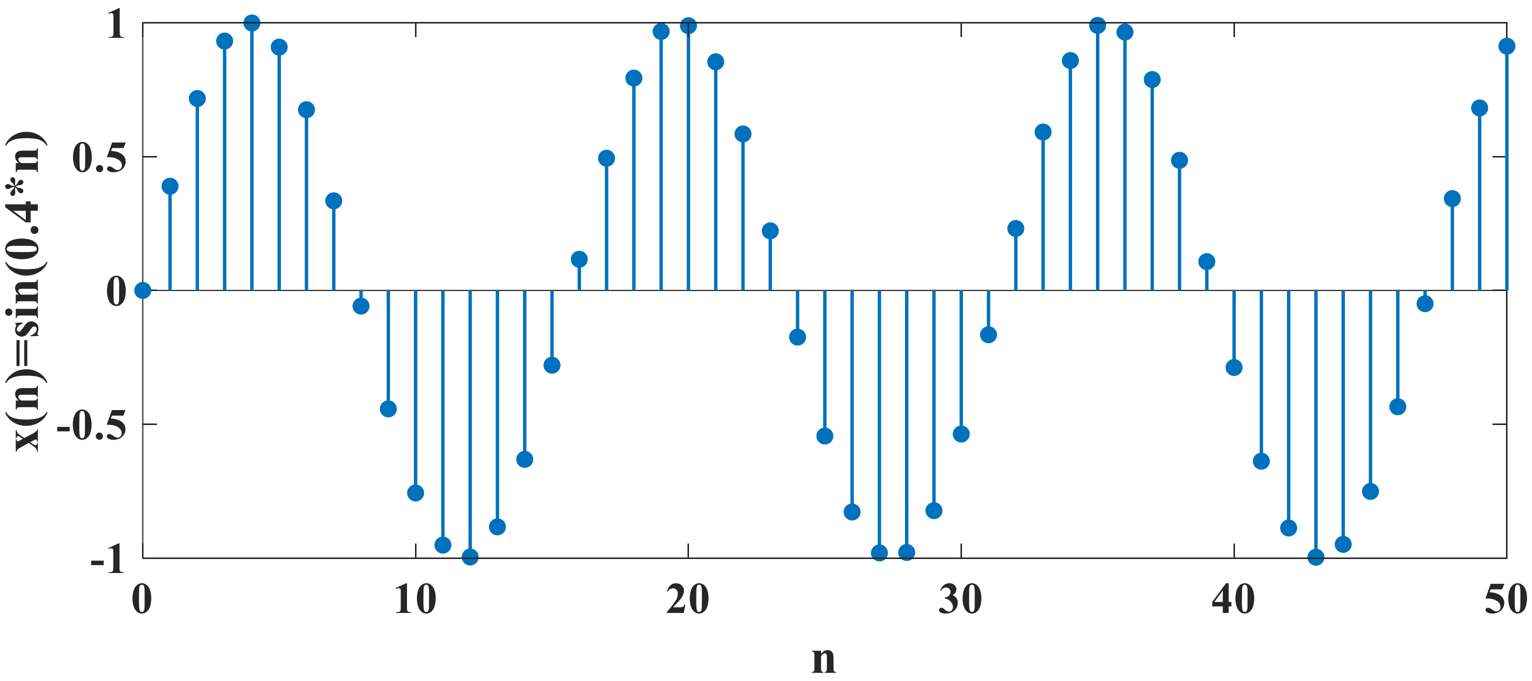
$\sin(0.4n)$ is a digital signal. The statement is (  )

A Ture

B False

提交

Sampling $x(t) = \sin(0.4t)$ yields $x(n) = \sin(0.4n)$。If $x(n)$ is a periodic signal, there exists an integer $N$ that satisfies $\sin(0.4n)$= $\sin[\,0.4(n+N)]$. On the other hand, one can derive $0.4N = 2k\pi$， i.e., $N = 5k\pi$ ($k$ is an integer), which is an irrational number and cannot be an integer. Therefore, $x(n)$ is a **non-periodic** signal. The amplitude cannot be represented by a finite set of discrete levels. $x(n)$ **is a discrete-time signal, but NOT a digital signal**.



## 4. Energy signals and power signals

**Energy of real-valued signal $f(t)$:** $E = \lim_{T_0 \to \infty} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} f^2(t)\,\mathrm{d}t$

**Average power of signal $f(t)$:** $P = \lim_{T_0 \to \infty} \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} f^2(t)\,\mathrm{d}t$

**Energy signal:**

$0 < E < \infty$ (with finite value), $P = 0$

**Power signal:**

$0 < P < \infty$ (with finite value), $E = \infty$

① Generally, periodic signals are power signals.

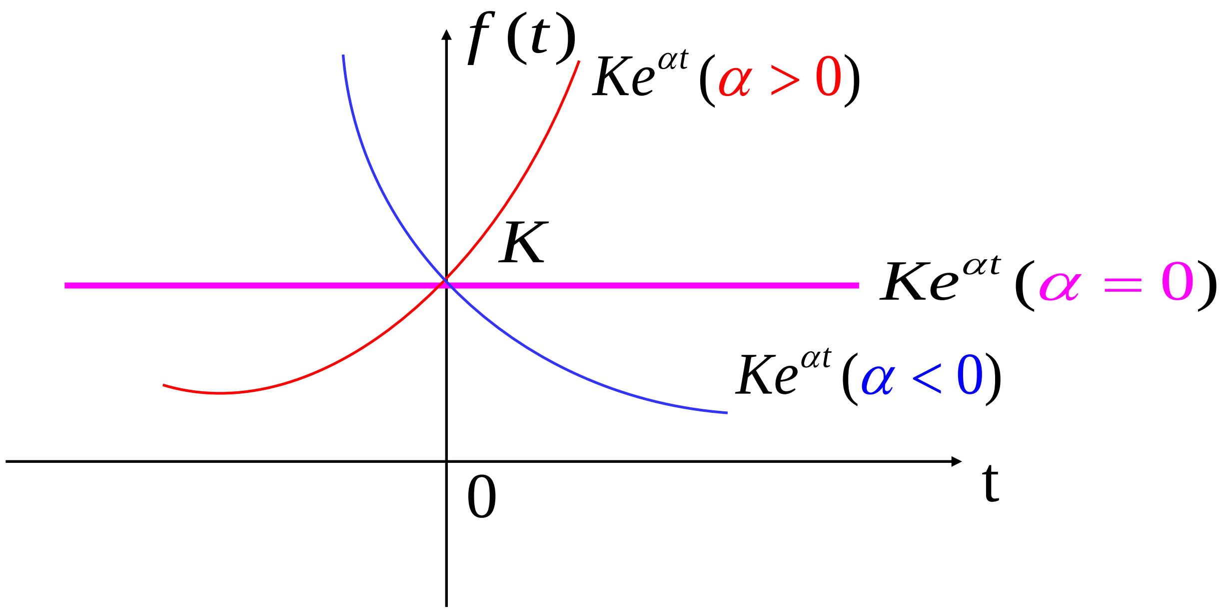
② Non-periodic signals with values within a limited interval are energy signals.

③ Some non-periodic signals are neither energy signals nor power signals.

## 1.2.2 Typical Signals

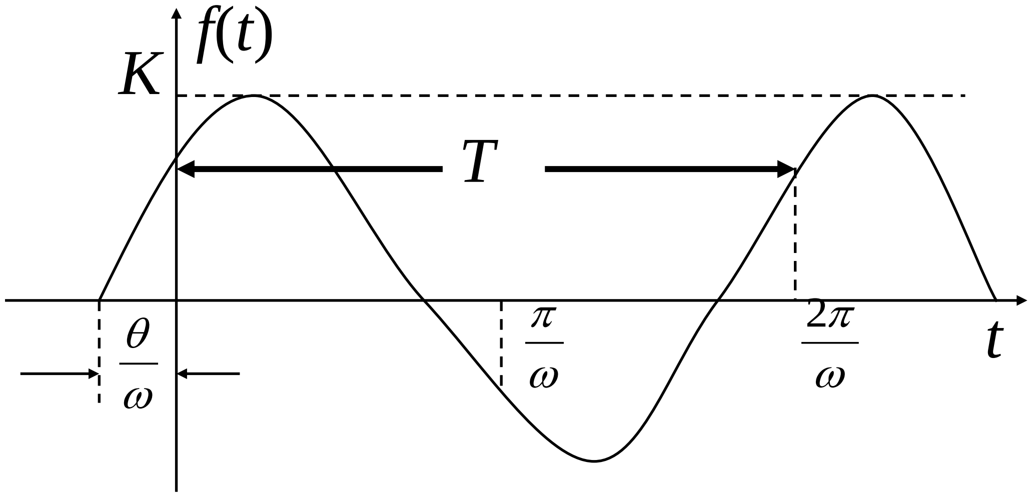
### 1. Exponential Signal

Expression: $f(t) = Ke^{\alpha t}$




### 2. Sinusoidal Signal

$$f(t) = K\sin(\omega t + \theta)$$



## 3. Complex Exponential Signal

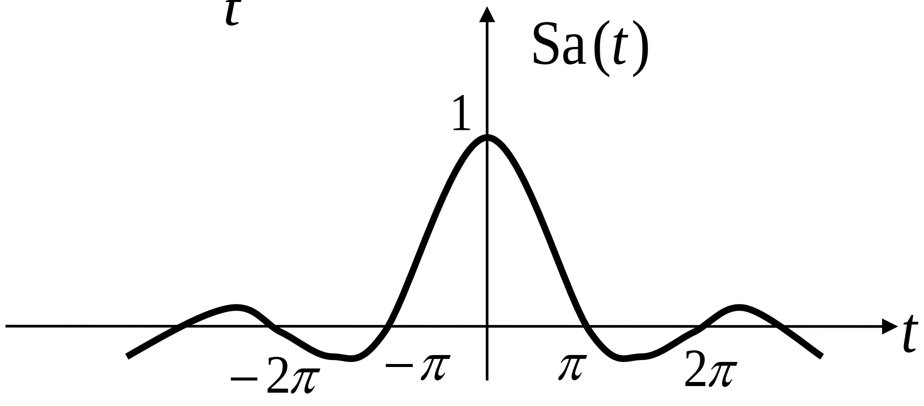
If the exponential factor of an exponential signal is a complex number, it is referred to as a complex exponential signal, expressed as:

$$f(t) = Ke^{st} = Ke^{(\sigma + j\omega)t} = Ke^{\sigma t}\cos\omega t + jKe^{\sigma t}\sin\omega t$$

## 4. Sampling function Sa(*t*) and Sinc(t)

The sampling function is defined as:

$$\mathrm{Sa}(t) = \frac{\sin t}{t}$$




An alternative function is defined as:

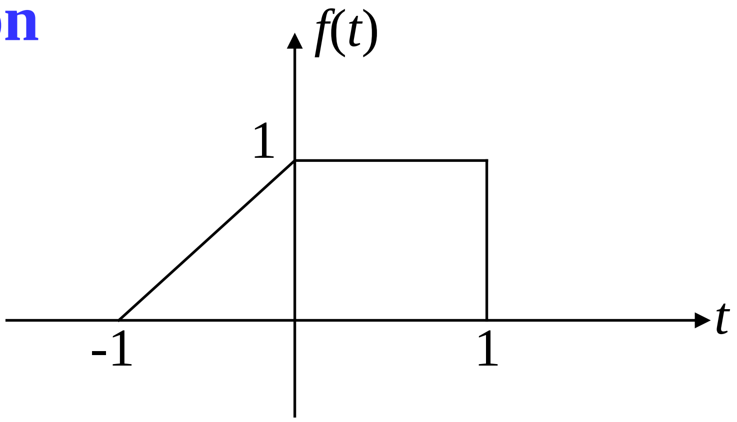
$$\mathrm{Sinc}(t) = \frac{\sin(\pi t)}{\pi t}$$

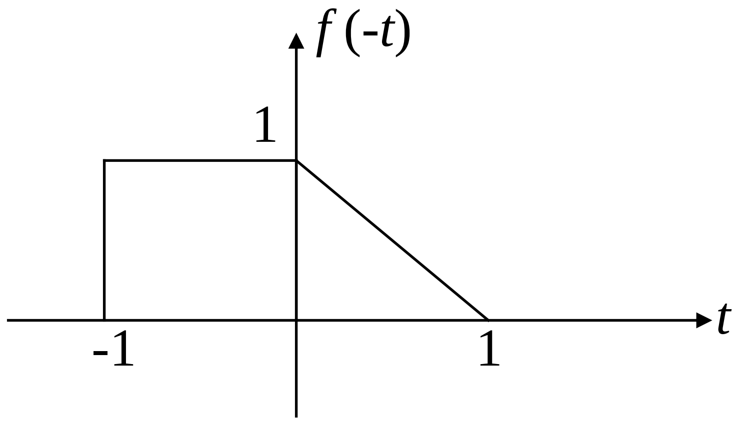
## 1.3.1 Time Operations-Reverse, Shifting, and Scaling

### 1. Reverse Operation

$$f(t) \to f(-t)$$

**Reverse against the *t* = 0 axis.**



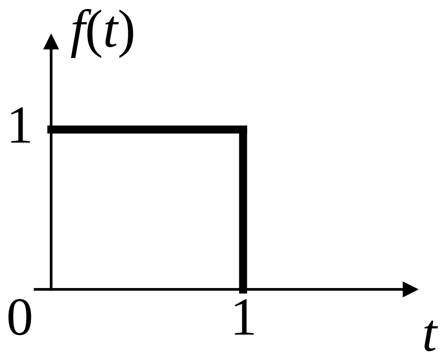



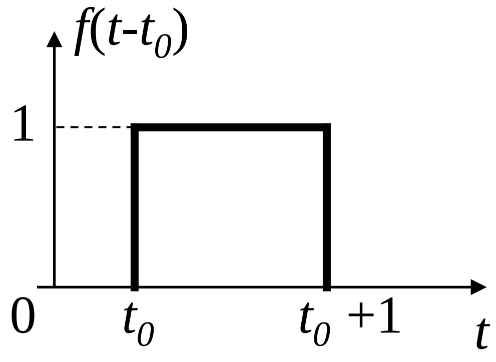

### 2. Time Shifting Operation

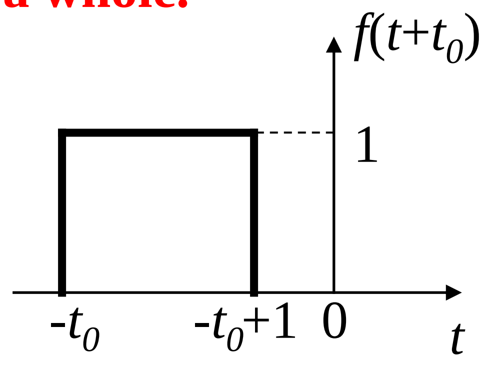
$$f(t) \to f(t - t_0)$$

**$t_0$>0时，$f(t)$ shifts to the right by $t_0$ as a whole.**

**$t_0$<0时，$f(t)$ shifts to the left by $t_0$ as a whole.**



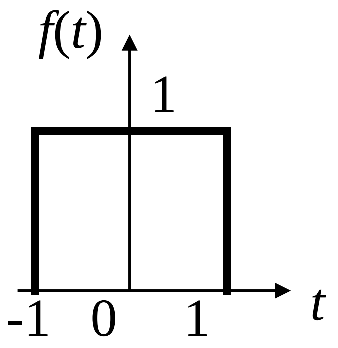
## 3. Scaling Operation

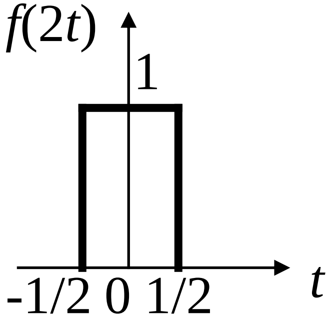
$f(t) \to f(2t)$

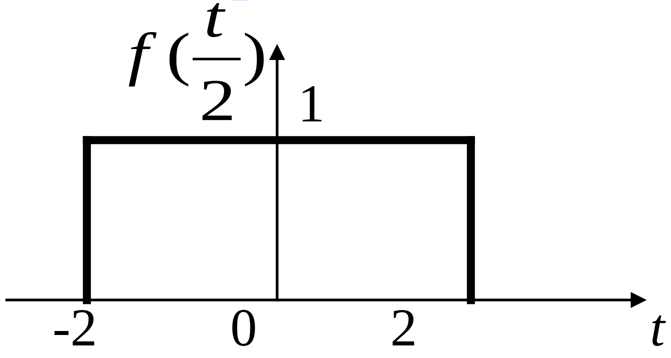
**Compression**

$f(t) \to f(\frac{t}{2})$

**Expansion**




$f(t) \to f(at)$

**Compression**: If $|a| > 1$, the waveform is compressed along the time axis.

**Expansion**: If $0 < |a| < 1$, the waveform is expanded along the time axis.

## 1.3.2 Differentiation and Integration Operations of Signals

### 1. Differentiation Operation

The derivative of signal $f(t)$ is still a signal, represented by $f'(t)$ ,which represents the rate of change of the signal with time. The differentiation operation highlights the changing part of the signal.

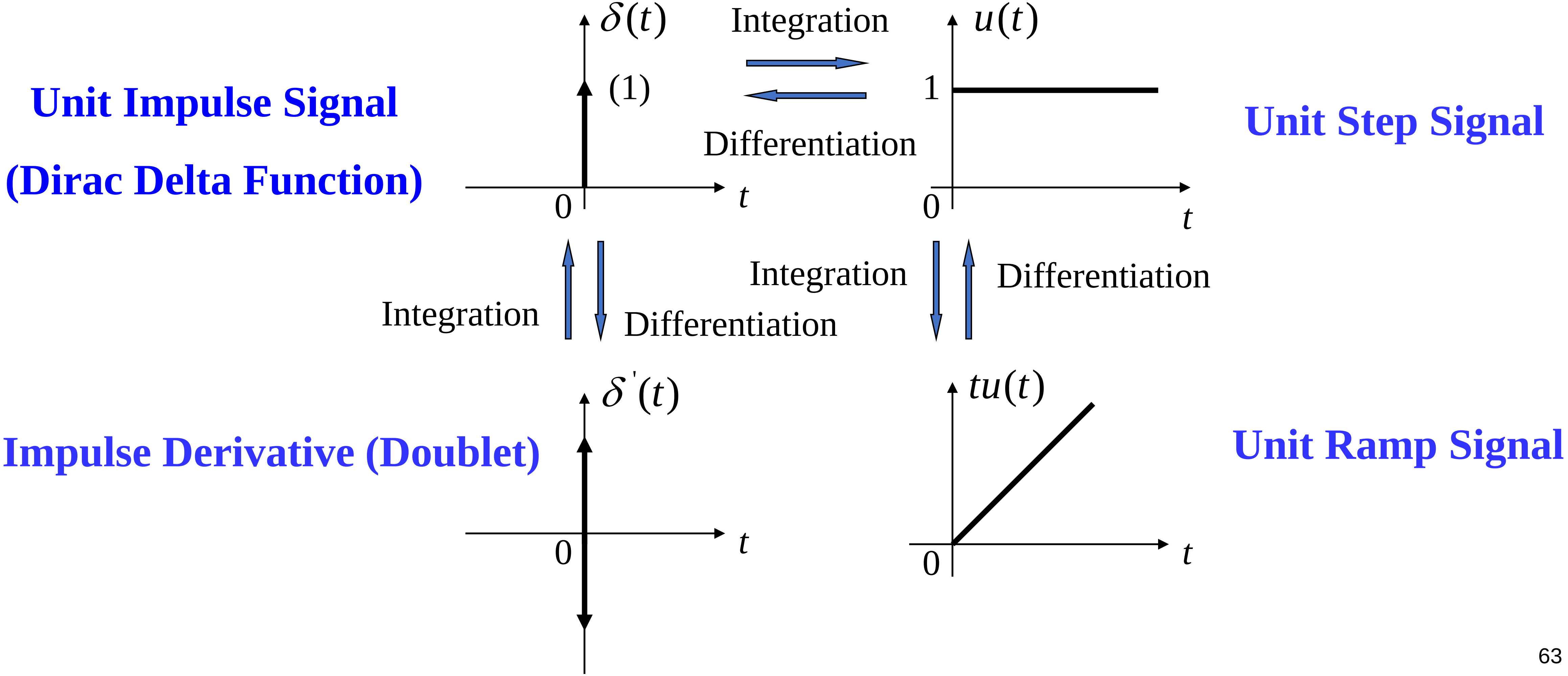
### 2. Integration Operation

- The integral of signal $f(t)$, denoted by $\int_{-\infty}^{t} f(\tau)d\tau$ or $f^{(-1)}(t)$ , is still a signal. Its value at any time is equal to the area enclosed by $f$(t) and the time axis within the interval from -∞ to t.

- The integration operation smooths the abrupt parts of the signal and can reduce the impact of glitches (noise).

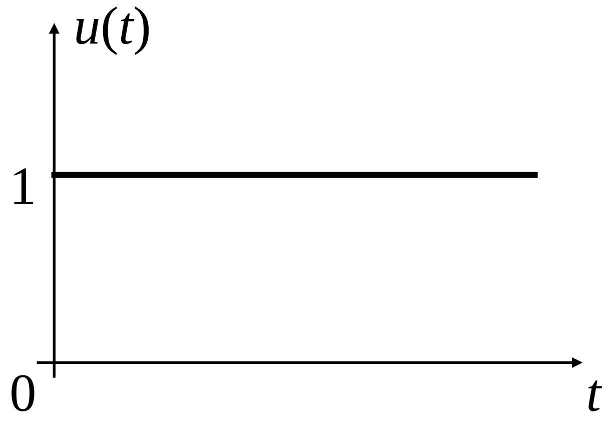
Singular signals: The signals with discontinuities in themselves or their derivatives/integrals.

Relationship between $tu(t)$, $u(t)$, $\delta(t)$, and $\delta'(t)$:

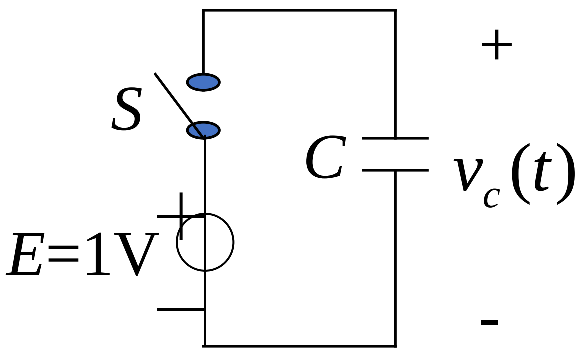


## Unit Step Signal

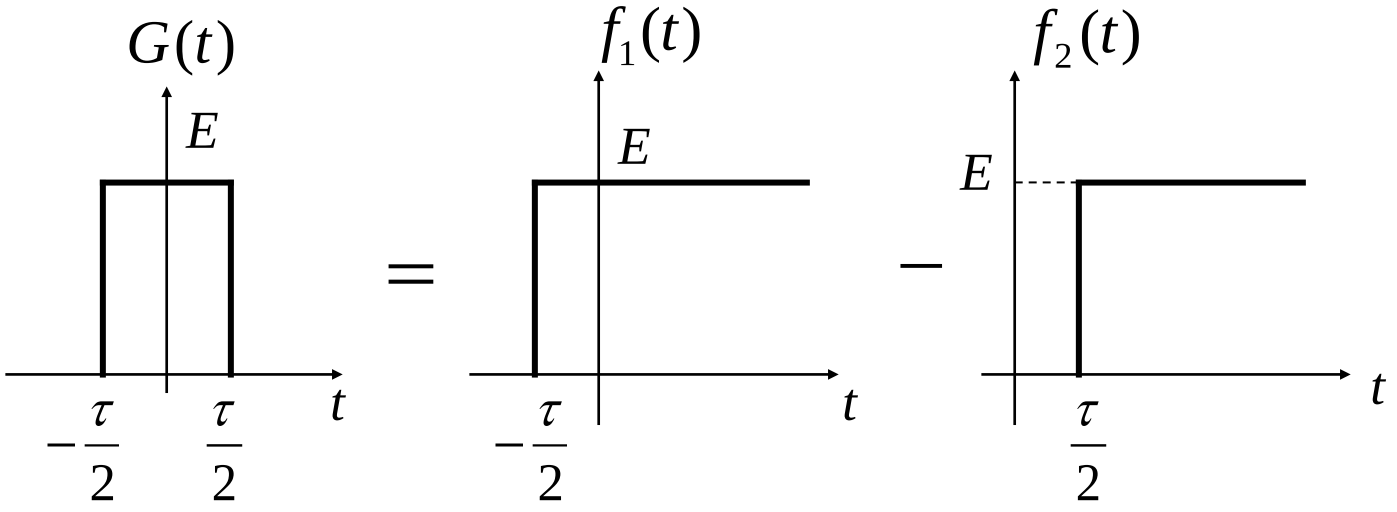
$$u(t) = \begin{cases} 0, (t < 0) \\ 1, (t > 0) \end{cases}$$




Example: The voltage across the capacitor C when S is closed at t = 0 is a unit step signal.



**Properties of *u*(*t*):** Unilateral characteristic, i.e., $f(t)u(t)=\begin{cases}0 & t<0\\ f(t) & t>0\end{cases}$



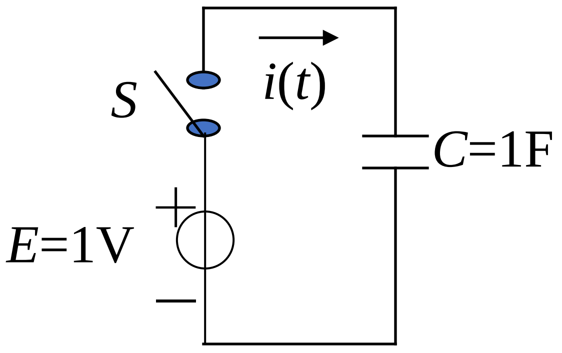

Since $f_1(t)=Eu(t+\frac{\tau}{2})$, $f_2(t)=Eu(t-\frac{\tau}{2})$,

the rectangular pulse *G*(*t*) can be expressed as:

$$G(t)=f_1(t)-f_2(t)=E[u(t+\frac{\tau}{2})-u(t-\frac{\tau}{2})]$$

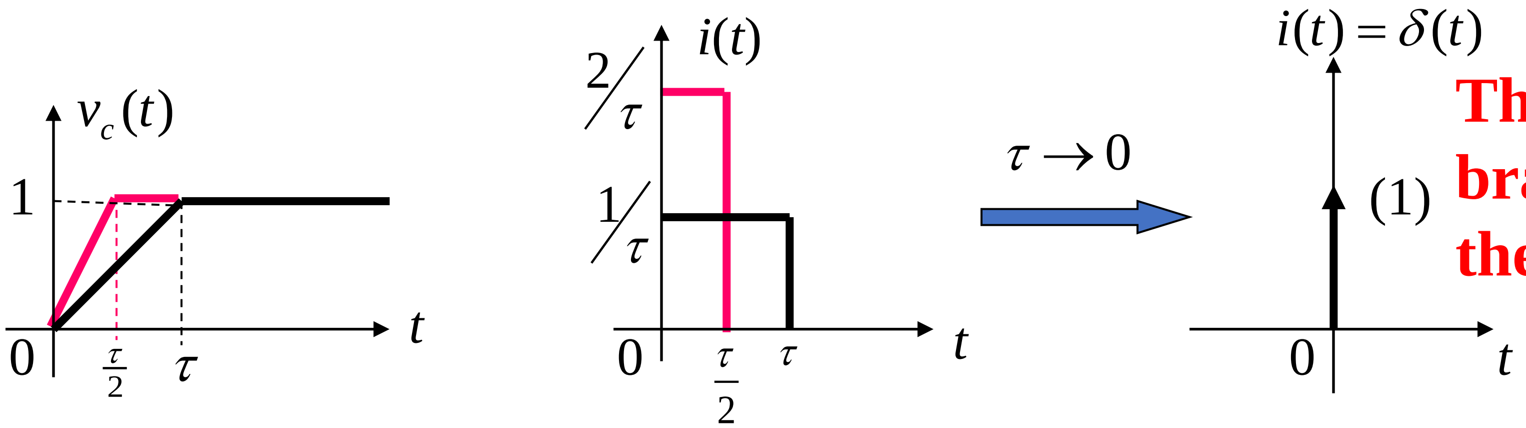
## Unit Impulse Signal (Dirac Delta Function)

Physical meaning of the unit impulse signal (Dirac delta function, $\delta(t)$) with the following example：




Assume that S, E, and C in the figure are all ideal components (internal resistance = 0). Find current $i(t)$ when S is closed at t = 0.

$$i(t) = C\frac{dv_C(t)}{dt}$$

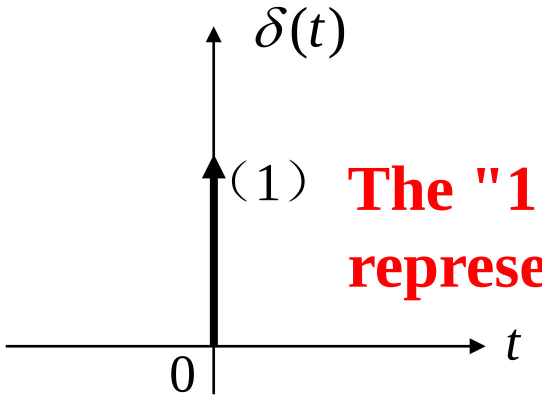



**The "1" in the bracket represents the strength.**

**Definition of** $\delta(t)$

**1）By Expression**

$$\begin{cases} \delta(t) = 0 \quad (t \neq 0) \\ \int_{-\infty}^{\infty} \delta(t)dt = 1 \end{cases}$$




**The "1" in the bracket represents the strength.**

**2）** $\delta(t)$ is an even function, i.e., $\delta(t) = \delta(-t)$

**3）** $\int_{-\infty}^{t} \delta(\tau)d\tau = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases} = u(t)$

$$\int_{-\infty}^{t} \delta(\tau - t_0)d\tau = u(t - t_0)$$

## Properties of the Unit Impulse Signal

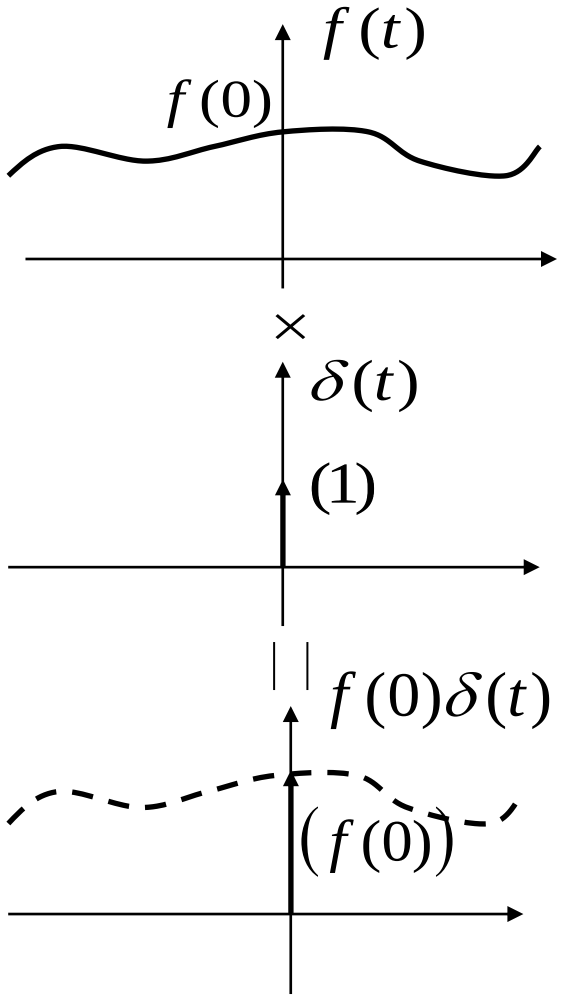
### 1）Sampling

$$f(t)\delta(t) = f(0)\delta(t) \qquad (1)$$

$$\int_{-\infty}^{\infty} \delta(t) f(t) dt = f(0) \int_{-\infty}^{\infty} \delta(t) dt = f(0) \qquad (2)$$

$$f(t)\delta(t-t_0) = f(t_0)\delta(t-t_0) \qquad (3)$$

$$\int_{-\infty}^{\infty} \delta(t-t_0) f(t) dt = f(t_0) \qquad (4)$$




Combining formulas (2) and (4), we can draw the following conclusion:

**The impulse function can extract (sample) the function value at the position where the impulse is located.**

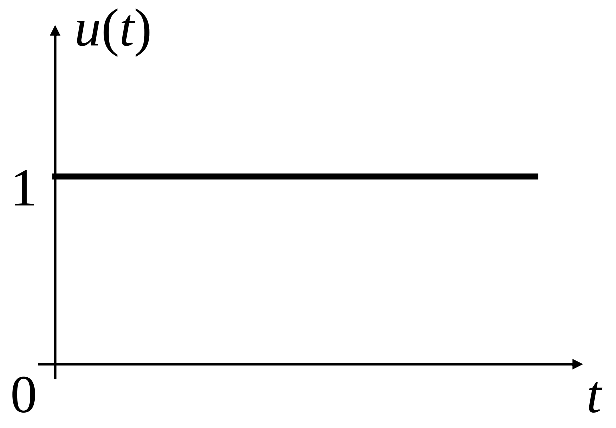
Relationship between $u(t)$ and $\delta(t)$

$$\int_{-\infty}^{t} \delta(\tau)d\tau = u(t)$$

$$\frac{d}{dt}u(t) = \delta(t)$$

$$\int_{-\infty}^{t} \delta(\tau - t_0)d\tau = u(t - t_0)$$

$$\frac{d}{dt}u(t - t_0) = \delta(t - t_0)$$



## 单选题 1分

设置

$$f(t)=\frac{\mathrm{d}}{\mathrm{d}t}\left[\mathrm{e}^{-t}\delta(t)\right]=(\quad)$$

A $\delta(t)$

B $\delta'(t)$

C $e^{-t}\delta'(t)$

D $-e^{-t}\delta'(t)$

提交

## Contents of This Lecture



## Learning Objectives of This Lecture

1. Master the method of decomposing any signal into pulse components;

2. Understand different classification methods of systems;

3. Be able to accurately judge the **linearity, time-invariance, and causality** of a system;

4. Initially understand the analysis methods of systems.

## 1.5.1 Decomposition of Any Signal into the Sum of Even and Odd Components

**Even component:** $f_e(t) = f_e(-t)$

**Odd component:** $f_o(t) = -f_o(-t)$

Any signal can be decomposed into the sum of an even component and an odd component, i.e.,

$$f(t) = f_e(t) + f_o(t) \qquad (1)$$

$$f(-t) = f_e(t) - f_o(t) \qquad (2)$$

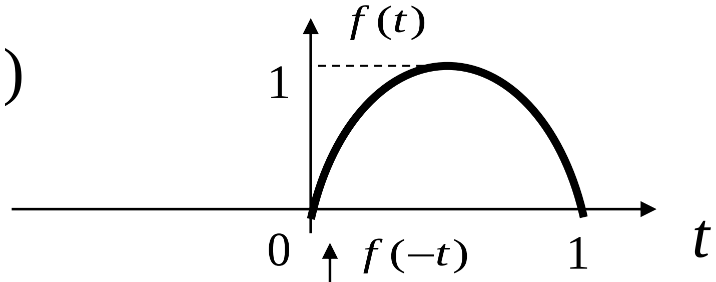
$$(1)+(2): \quad f_e(t) = \frac{1}{2}[f(t) + f(-t)]$$

$$(1)-(2): \quad f_o(t) = \frac{1}{2}[f(t) - f(-t)]$$

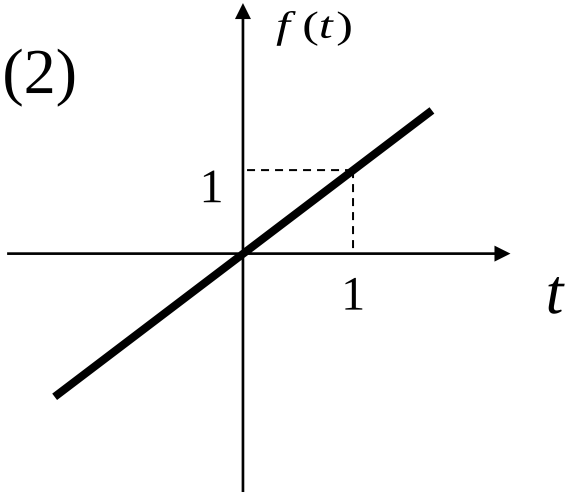
# 1.5 Decomposition of Signals

**Example 1-9:** Find the even and odd components of the signals shown in the figures.

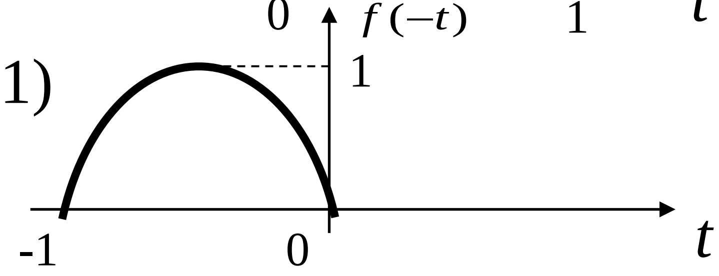
(1)

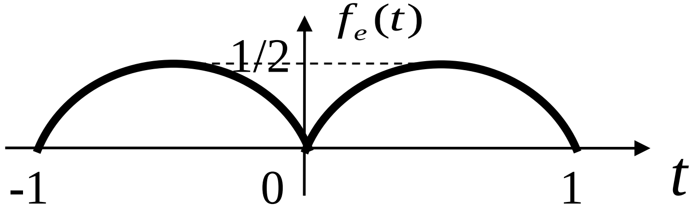


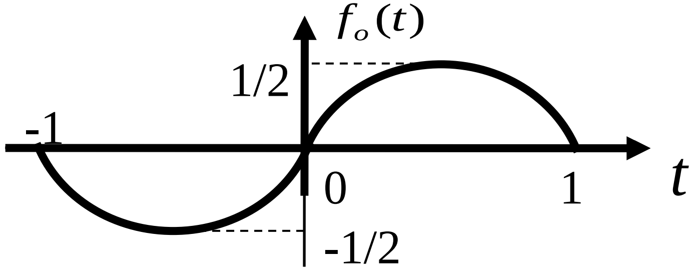

(2)




**Solution:** (1)



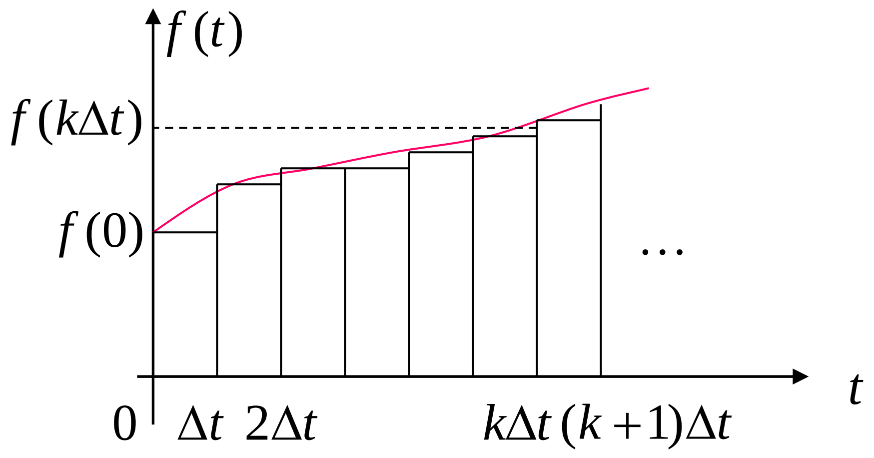

(2) This is an odd signal, so

$$f_o(t) = f(t)$$

$$f_e(t) = 0$$

### 1.5.2 Decomposition of Any Signal into Pulse Components

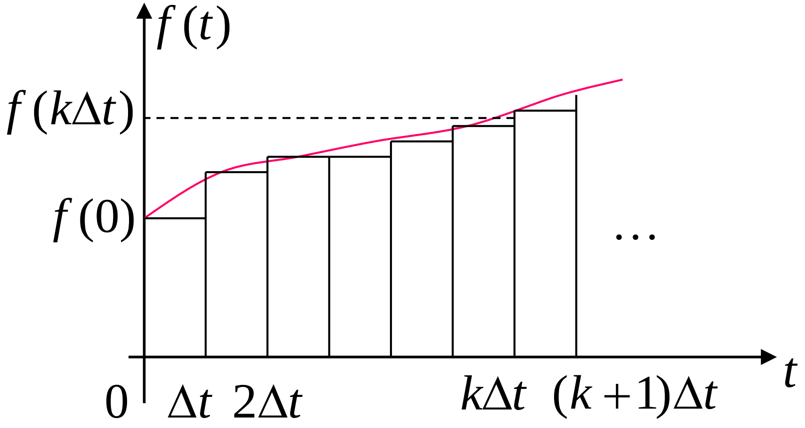
A signal can be approximately decomposed into the sum of many pulse components. A common decomposition is the decomposition into **rectangular narrow pulse components**, and the limit of the combination of narrow pulses is the **superposition of impulse signals**.




When $t = 0$, the first rectangular pulse is

$$f(0)[u(t)-u(t-\Delta t)]$$

$$\lim_{\Delta t\to 0}\frac{f(0)[u(t)-u(t-\Delta t)]}{\Delta t}\Delta t=\lim_{\Delta t\to 0} f(0)\delta(t)\Delta t$$

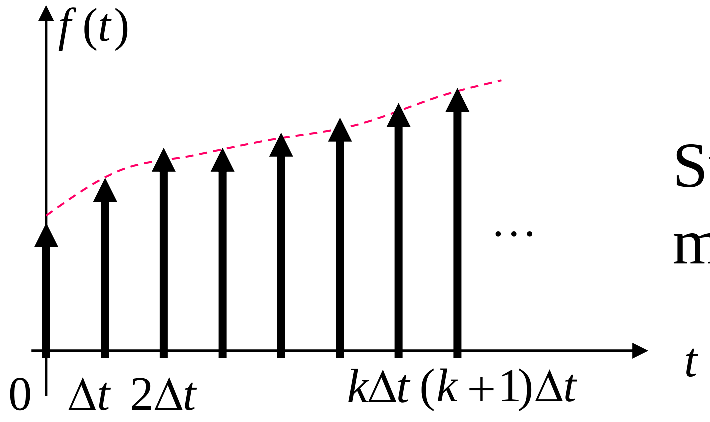



When $t = k\Delta t$ , the ($k$+1)-th rectangular pulse is

$$f(k\Delta t)\{u(t-k\Delta t)-u[t-(k+1)\Delta t]\}$$

$$\lim_{\Delta t\to 0} f(k\Delta t)\frac{\{u(t-k\Delta t)-u[t-(k+1)\Delta t]\}}{\Delta t}\Delta t$$

$$=\lim_{\Delta t\to 0} f(k\Delta t)\delta(t-k\Delta t)\Delta t$$




Superimposing the 0 to $n$ rectangular pulses mentioned above, we get the expression of $f$(t), *i.e.*,

$$f(t)=\lim_{\Delta t\to 0}\sum_{k=0}^{n} f(k\Delta t)\delta(t-k\Delta t)\Delta t$$

When $\Delta t \to 0,\ \Delta t \to d\tau, k\Delta t \to \tau, \lim_{\Delta t\to 0}\sum_{k=0}^{n} \to \int_{0^-}^{t}$

$$\boxed{f(t)=\int_{0^-}^{t} f(\tau)\delta(t-\tau)d\tau}$$

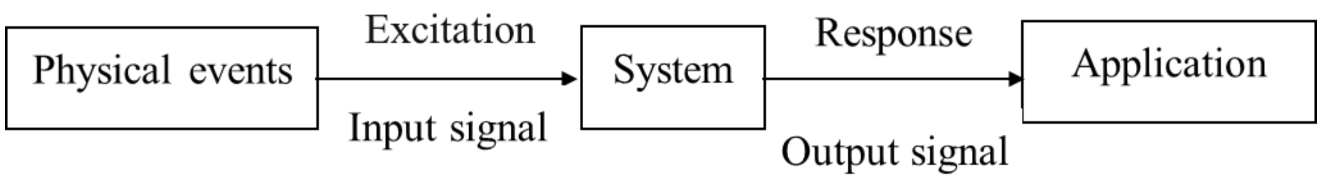
## 1.5.3 Decomposition of a Signal into Orthogonal Function Components

If a signal is represented by an orthogonal function set, then each component composing the signal is mutually orthogonal.

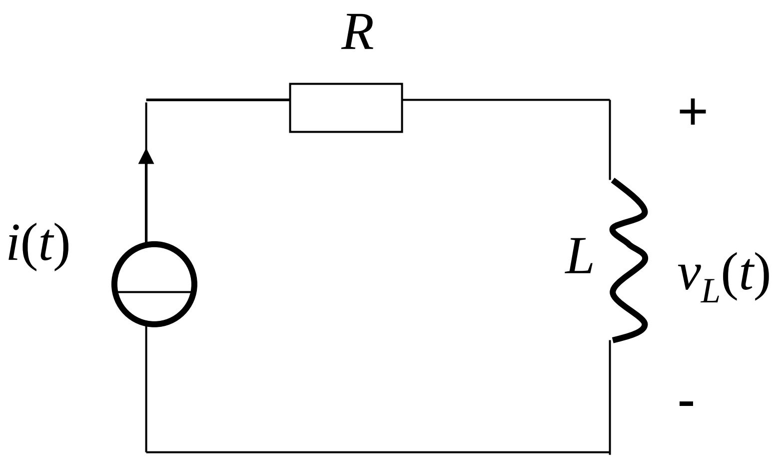
For example, the trigonometric function set composed of sine and cosine signals of various harmonics is an orthogonal function set. A periodic signal f(t) can be represented by the linear combination of these trigonometric functions as long as it satisfies the Dirichlet conditions, which is called the trigonometric form of the Fourier series of f(t). Similarly, f(t) can also be expanded into the exponential form of the Fourier series. (Content of Chapter 3)

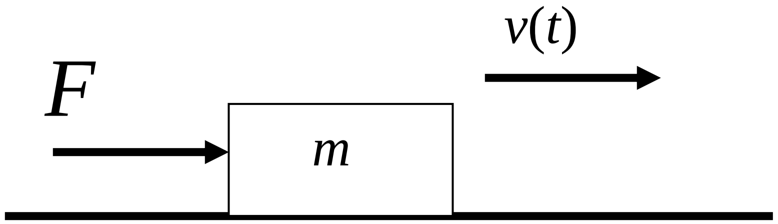
## Definition of a System

- A system is an integrated whole composed of several interrelated and interacting things, with certain or specific functions. Examples include communication systems and radar systems.
- The concept of a system is not only applicable to various fields of natural sciences but also to social sciences, such as political structures and economic organizations.
- A common feature of systems in many different fields is that all systems respond to the signals applied to them (i.e., the **input signals** of the system, also called **excitations**) and generate other signals (i.e., the **output signals** of the system, also called **responses**). The function of a system is reflected in what kind of output signal is generated by a certain input signal.
- Each system has its own **mathematical model**.



## 1.6.1 Mathematical Models of Systems



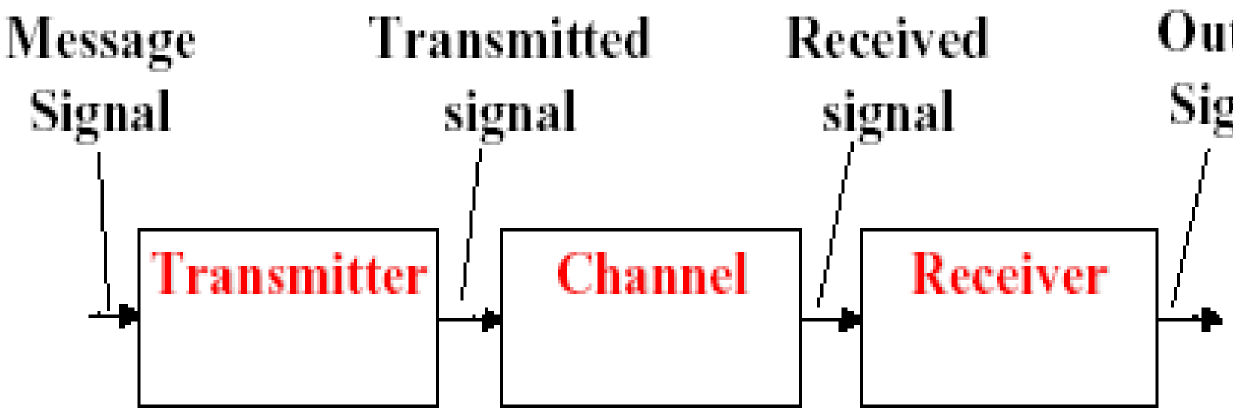




$$v_L(t) = L\frac{di(t)}{dt}$$

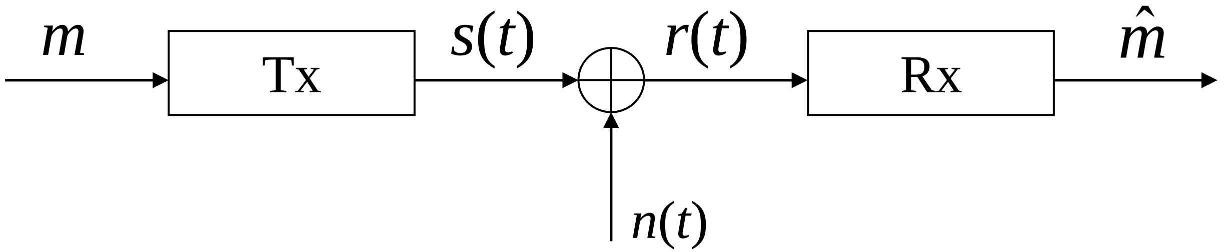
$$F = ma = m\frac{dv(t)}{dt}$$

Two different systems may have the same mathematical model, and even physical systems and non-physical systems may have the same mathematical model. Systems with the same mathematical model are called similar systems.

## Communication System

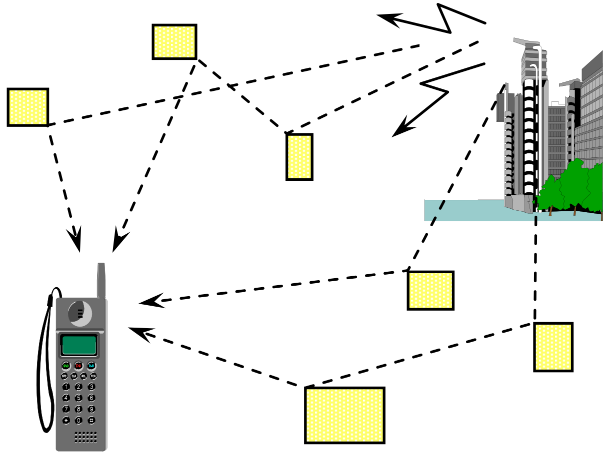



**Additive White Gaussian Noise (AWGN) Channel Model**

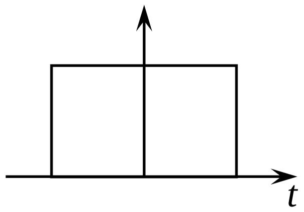



$$r(t) = s(t) + n(t)$$

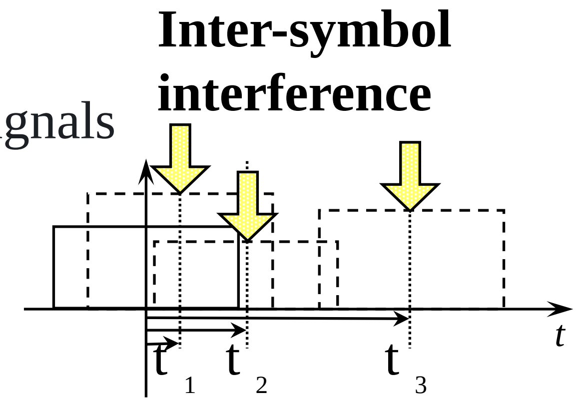
## Multipath Channel Model



Transmitted signal




Received signals



## 1.6.2 Classification of Systems

### (1) Continuous-Time Systems and Discrete-Time Systems

•The mathematical model of a continuous-time system is a differential equation.
•The mathematical model of a discrete-time system is a difference equation.

### (2) Instantaneous Systems (Memoryless Systems) and Dynamic Systems (Memory Systems)

•The mathematical model of an instantaneous system is an algebraic equation, such as a resistor circuit.
•The mathematical model of a dynamic system is a differential equation or a difference equation, such as an RC or RL circuit.

### (3) Lumped Parameter Systems and Distributed Parameter Systems

•The mathematical model of a lumped parameter system is an ordinary differential equation.
•The mathematical model of a distributed parameter system is a partial differential equation.

## (4) Linear Systems and Nonlinear Systems

- A system with superposition and homogeneity (also called scalability) is called a linear system.
- A system that does not satisfy superposition or homogeneity is called a nonlinear system.

## (5) Time-Varying Systems and Time-Invariant Systems

- Time-varying system: The parameters of the system change with time.
- Time-invariant system: The parameters of the system do not change with time.

## (6) Single-Input Single-Output (SISO) Systems and Multiple-Input Multiple-Output (MIMO) Systems

- SISO system: Only accepts one excitation and generates one response.
- MIMO system: The system has more than one excitation signal and more than one response signal. For example, the massive MIMO system in 5G.

## (7) Invertible Systems and Non-Invertible Systems

•Invertible system: Different excitations generate different responses.

•Non-invertible system: Different excitations generate the same response.

•For each invertible system, there exists an “inverse system”. When the original system and its inverse system are cascaded, the output signal is the same as the input signal.

- Examples:

- An invertible system: $r(t) = 3e(t)$

  Its inverse system: $r(t) = e(t)/3$

- A non-invertible system: $r(t) = e^2(t)$

When the excitations are e(t) = 1 and e(t) = -1, the response r(t) is 1 in both cases. That is, different excitations generate the same response, so it is a non-invertible system.

# 1.7 Linear Time-Invariant Systems

## 1.7.1 Linearity

Linearity includes superposition and homogeneity (also called scalability).

### 1. Superposition

$$\begin{matrix} x_1(t) \\ x_2(t) \end{matrix} \longrightarrow \boxed{\textbf{System}} \longrightarrow \begin{matrix} y_1(t) \\ y_2(t) \end{matrix}$$

If $\quad x(t) = x_1(t) + x_2(t) \quad , \qquad y(t) = y_1(t) + y_2(t) \quad ,$

the system is said to satisfy superposition.

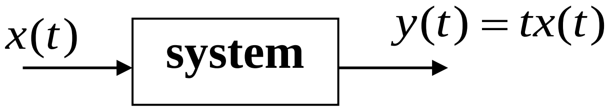
### 2. Homogeneity

If $\quad x(t) = ax_i(t) \quad , \qquad y(t) = ay_i(t) \quad ,$

the system is said to satisfy homogeneity.

A system that satisfies both superposition and homogeneity is called a **linear system**.

**Example 1-10**: Assume the input-output relationship of a system is $y(t)=tx(t)$ . Determine whether the system is linear.




**Solution:**

Check superposition:

$$y_1(t)=tx_1(t) \qquad y_2(t)=tx_2(t) \qquad x(t)=x_1(t)+x_2(t)$$

$$y(t)=t[x_1(t)+x_2(t)]=tx_1(t)+tx_2(t)=y_1(t)+y_2(t)$$

Superposition is satisfied.

Check homogeneity:

$$x(t)=ax_1(t) \qquad y(t)=tx(t)=atx_1(t)=ay_1(t)$$

Homogeneity is satisfied.

$$\boxed{x(t)=\sum_{k=1}^{N}a_k x_k(t) \qquad y(t)=\sum_{k=1}^{N}a_k y_k(t)}$$

Hence, the system is **linear**.

单选题 1分

设置

Assume the input-output relationship of a system is $y(t) = ax(t) + b \ (b \neq 0)$ , Determine whether the system is linear.

A Yes

B No

提交

**Example 1-11**: Assume the input-output relationship of a system is $y(t)=ax(t)+b\ \ (b\neq 0)$ , Determine whether the system is linear.

**Solution**:

Check superposition:

$$y_1(t)=ax_1(t)+b \qquad y_2(t)=ax_2(t)+b$$

$$x(t)=x_1(t)+x_2(t)$$

$$y(t)=a[x_1(t)+x_2(t)]+b\neq y_1(t)+y_2(t)$$

Therefore, the system does not satisfy superposition.

Check homogeneity:

$$x(t)=cx_1(t) \qquad y(t)=ax(t)+b=acx_1(t)+b\neq cy_1(t)$$

Therefore, the system does not satisfy homogeneity.

The system is **not linear**.

For a linear system, if the input of the system is zero at all times, the output must be zero, i.e., zero input produces zero output.

$$x(t)=\sum_{k=1}^{N}a_k x_k(t)=\sum_{k=1}^{N}0\cdot a_k=0$$

$$y(t)=\sum_{k=1}^{N}a_k y_k(t)=\sum_{k=1}^{N}0\cdot a_k=0$$

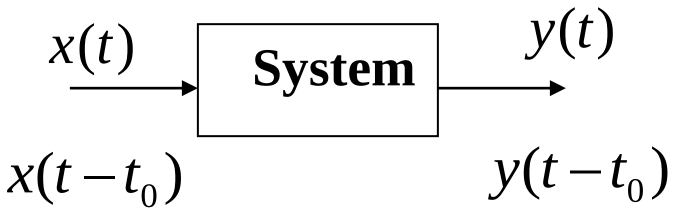
However, for the system,

$$y(t)=ax(t)+b=a\cdot 0+b=b$$

when the input $x(t)=0$, the output $y(t)=b\neq 0$. This non-zero output is called the **zero-input response** of the system.

## 1.7.2 Time-Invariance

Time-invariant system: The parameters of the system do not change with time; if the excitation is delayed, the response is delayed by the same amount.




**Example 1-12**: Determine whether the systems satisfying the following input-output relationships are time-varying: (1) $y(t)=tx(t)$ , (2) $y(t)=ax(t)+b$ .

**Solution:** (1) $y_1(t)=tx(t-t_0)\neq y(t-t_0)$

$$y(t-t_0)=(t-t_0)x(t-t_0)$$

However, $y$(t - $t_0$) = (t - $t_0$) $x$(t - $t_0$). Since $y_1$(t) ≠ $y$(t - $t_0$), the system is **time-varying**.

(2) $y_1(t)=ax(t-t_0)+b=y(t-t_0)$

So the system is **time-invariant**.

## 单选题 1分

设置

Assume the input-output relationship of a system is $y(t) = x(2t)$. Determine whether it is a time-invariant system.

A Time-Invariant

B Time-Varying

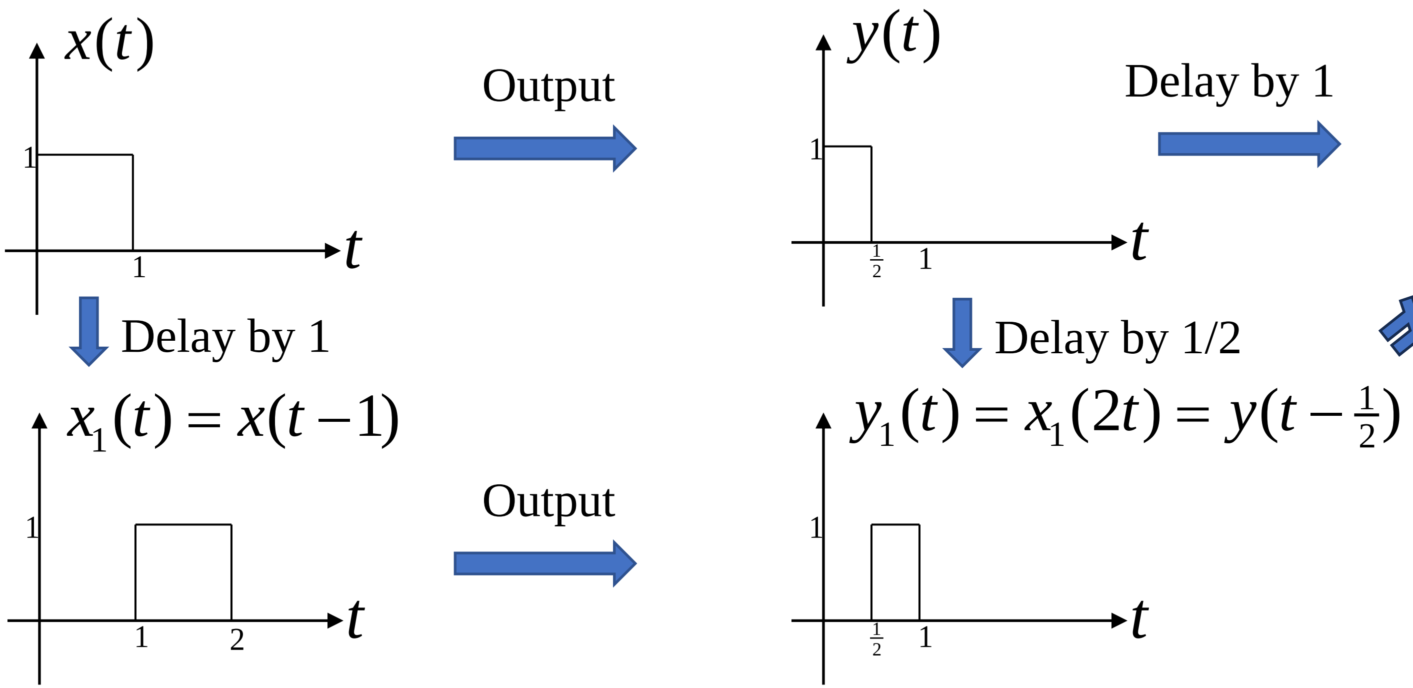
提交

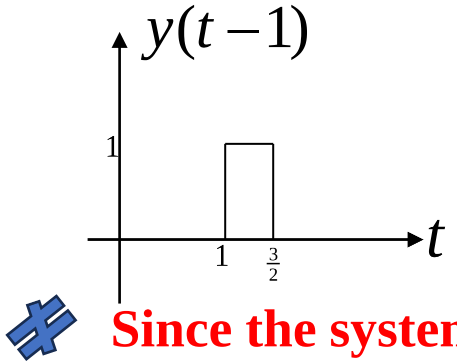
To determine whether a system satisfies a certain property, **we only need to find one example that does not satisfy it** to prove that the system does not have that property.

**Example 1-13**: Assume the input-output relationship of a system is $y(t) = x(2t)$. Determine whether it is a time-invariant system.

**Solution**: Define the delayed input of x(t) by $t_0$ as $x_1(t)=x(t-t_0)$, and the corresponding output is $y_1(t) = x_1(2t) = x(2t - t_0)$. However, the direct delay of y(t) by $t_0$ gives $y(t - t_0) = x[2(t - t_0)] = x(2t - 2t_0)$. The two are not equal, so the system is **time-varying**.

E.g.,




**Since the system compresses the input signal by 2 times in time, when the input signal is delayed by 1, the output signal is only delayed by 1/2 instead of 1.**

**Example 1-14**: Determine whether $y(t) = x(-t)$ is a time-invariant system.

**Solution**:

$$y(t) = x(-t)$$

$$x_1(t) = x(t - t_0)$$

$$y_1(t) = x(-t - t_0)$$

$$y(t - t_0) = x(-t + t_0) \neq y_1(t)$$

This system only reverses the excitation, i.e., multiplies $t$ in $x$ by -1.

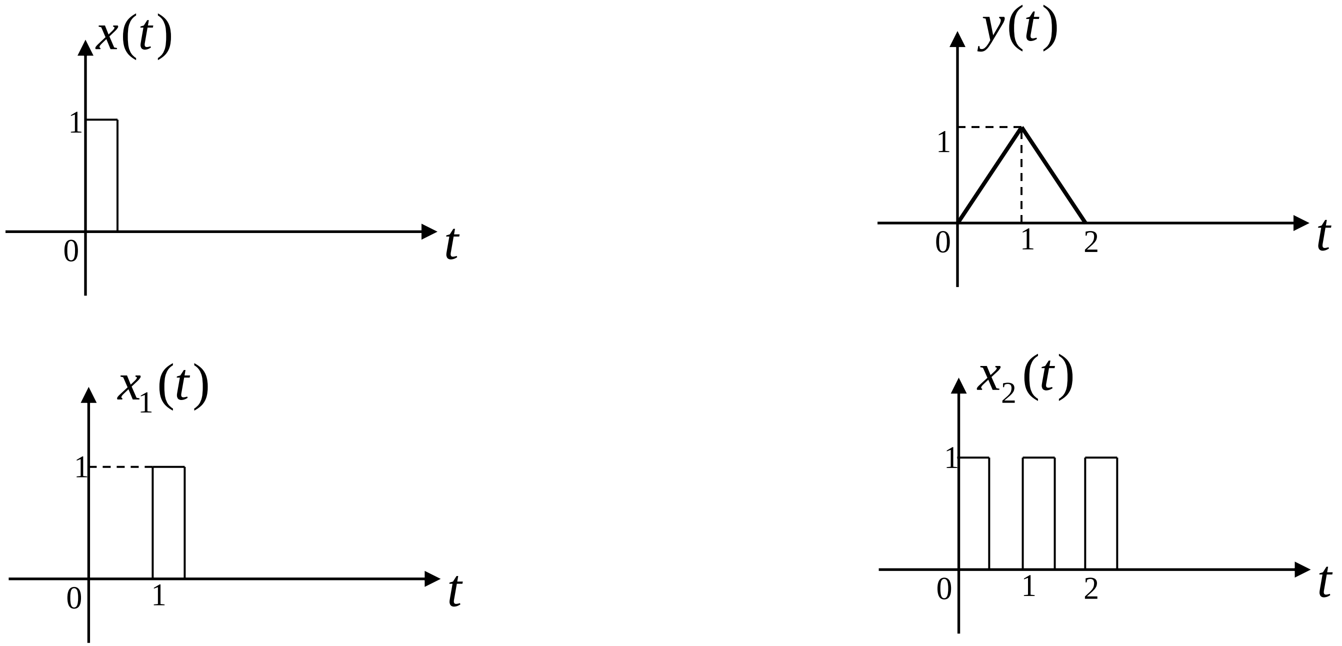
Since $y_1(t) \neq y(t - t_0)$, the system is **time-varying**.

Intuitive Judgment Method for Time-Invariance:
**If there is a time-varying coefficient in front of $x(\cdot)$ (e.g., $tx(t)$), or reverse (e.g., $x(-t)$), or scaling transformation (e.g., $x(2t)$), then the system is time-varying.**

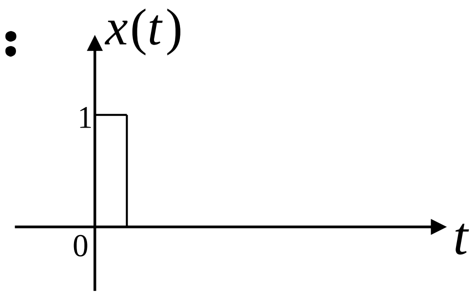
A system that satisfies both linearity and time-invariance is called a **Linear Time-Invariant (LTI)** system, which can be expressed as:

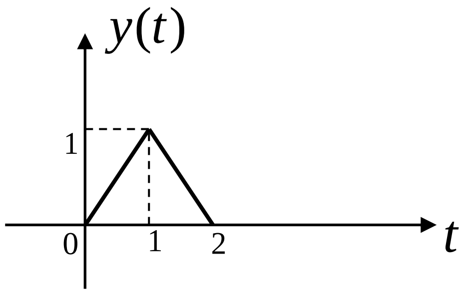
$$x(t)=\sum_{k=1}^{N}a_k x_k(t-t_k) \qquad y(t)=\sum_{k=1}^{N}a_k y_k(t-t_k)$$

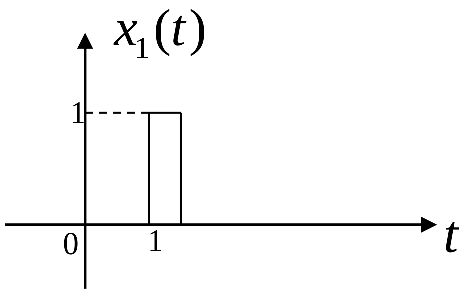
**Example 1-15**: Assume the input-output relationship of an LTI system is described in the figure. Draw the waveforms of outputs $y_1(t)$ and $y_2(t)$ when the inputs are $x_1(t)$ and $x_2(t)$ respectively.

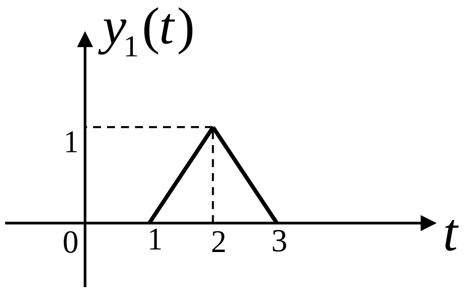


**Solution:**



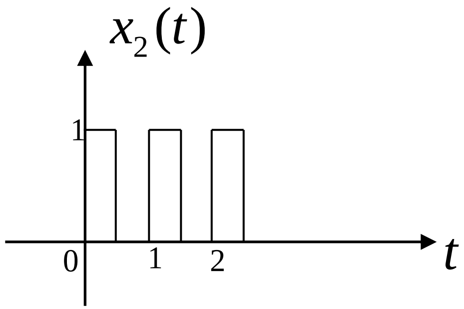



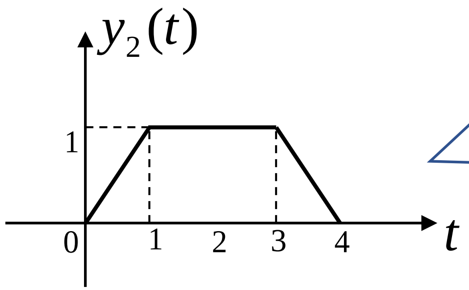



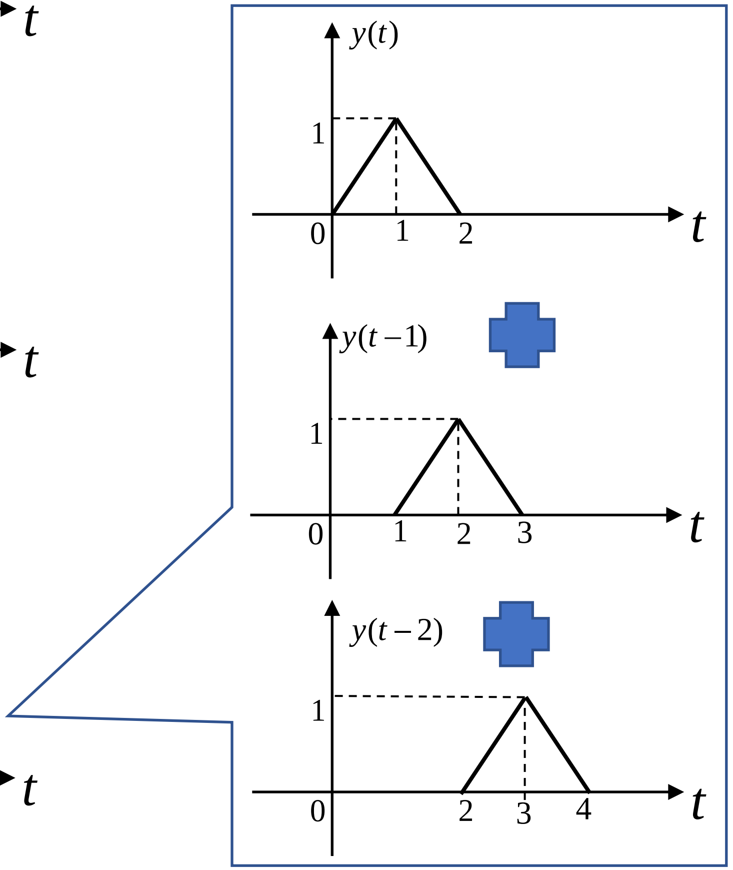




$$x_1(t) = x(t-1)$$

$$y_1(t) = y(t-1)$$




$$x_2(t) = x(t) + x(t-1) + x(t-2)$$

$$y_2(t) = y(t) + y(t-1) + y(t-2)$$

## 1.7.3 Differentiation and Integration Properties of Continuous-Time Systems

### 1. Differentiation Property

$$x(t) \longrightarrow \boxed{\text{LTI}} \longrightarrow y(t)$$

$$x'(t) \longrightarrow \boxed{\text{LTI}} \longrightarrow y'(t)$$

### 2. Integration Property

$$x(t) \longrightarrow \boxed{\text{LTI}} \longrightarrow y(t)$$

$$\int_{-\infty}^{t} x(\tau)d\tau \longrightarrow \boxed{\text{LTI}} \longrightarrow \int_{-\infty}^{t} y(\tau)d\tau$$

## 1.7.4 Causality

**A causal system is a system where the response at time $t_0$ depends only on the input at time $t \leq t_0$. Otherwise, it is a non-causal system.**

Examples:

Causal system: $r(t) = e(t-1)$ (delay system)

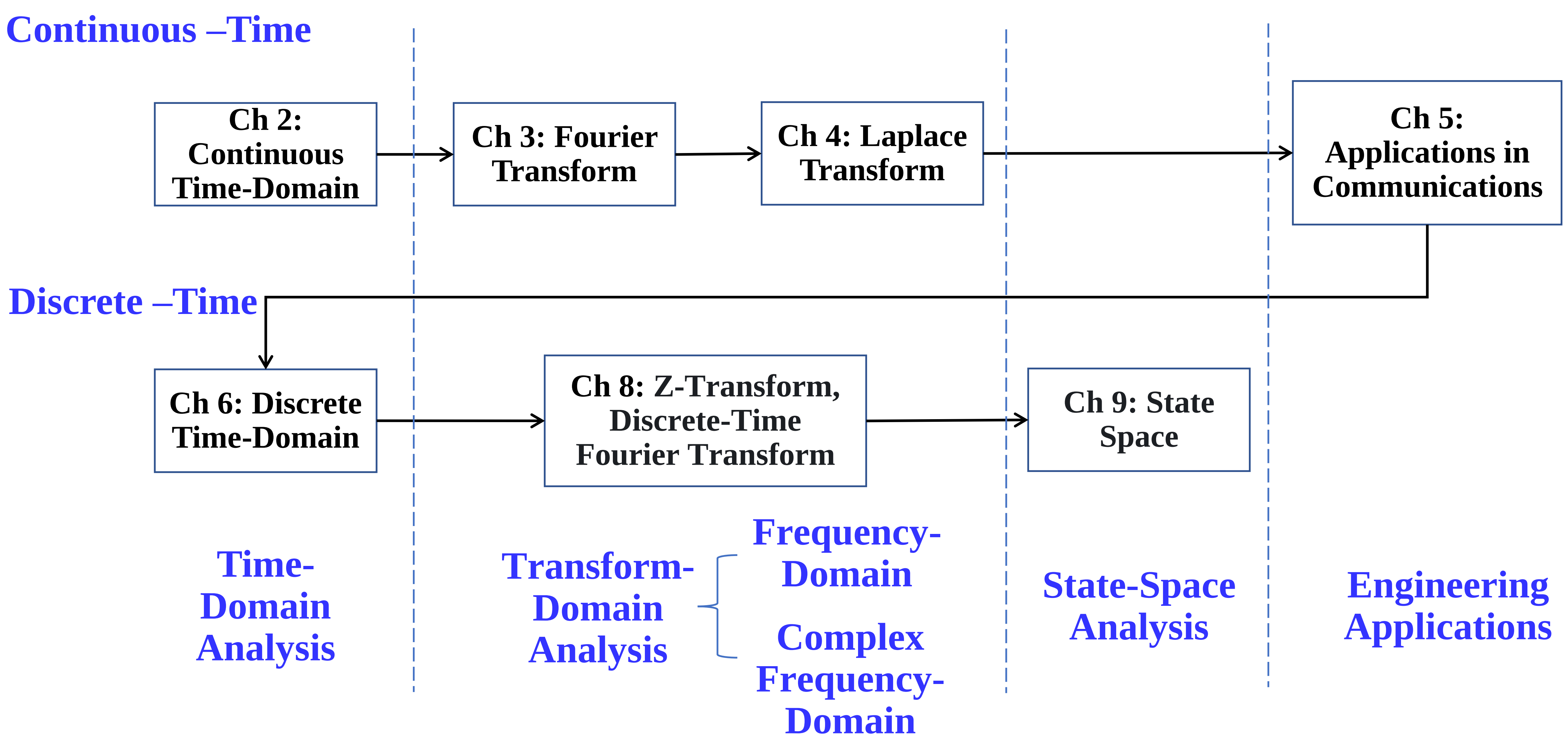
Non-causal system: $r(t) = e(t+1)$ (advance system)

(The response at time t = 0 is determined by the excitation at time t = 1, so it is a non-causal system.)

Non-causal system: $r(t) = e(2t)$ (time-compressed system)

Non-causal systems are common in fields such as speech signal processing, meteorology, stock market analysis, and demography.

# 1.8 Analysis Methods of LTI Systems



# 1.8 Analysis Methods of LTI Systems

## Classification by Mathematical Description

**Input-Output Analysis**: An n-th order differential (difference) equation, suitable for SISO systems (Chapters 2, 3, 4, 6 and 7)

**State Variable Analysis**: A system of *n* first-order differential equations, suitable for MIMO and nonlinear systems (Chapter 9)

## Classification by Solving Methods of System Mathematical Models

**Time-Domain Analysis**: Solve the response directly in the time domain without any transformation (Chapters 2 and 6)

**Transform-Domain Analysis**: Transform the time functions of signals and system models into functions in the corresponding transform domain, such as Fourier Transform (Chapters 3 and 5), Laplace Transform (Chapter 4), Z-Transform (Chapter 7), etc.

# Homework

- Homework: 1.64 (iii)(iv)(v)(a)(d)(f)(i)(k) in the textbook.
- Textbook: Simon Haykin and Barry Van Veen, Signals and Systems (Second Edition), John Wiley & Sons, 2002, 9780471164746.
- You do not need to submit the homework, but you are highly recommended to do the exercises. Refer to the solutions after you finish the homework.